GRAPH-BASED STOCHASTIC POWER-UCT: MONTE-CARLO GRAPH SEARCH WITH POWER MEAN ESTIMATION

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ABSTRACT

Tree-based Monte-Carlo Tree Search (MCTS) duplicates the same state when it is reached through different trajectories, which can waste simulations in stochastic MDPs. We introduce Graph-Based Stochastic-Power-UCT (GS-Power-UCT), which shares states reached at the same planning depth while keeping separate values for states reached at different depths. This design applies to general stochastic MDPs, including problems with cycles. We prove that for a fixed planning horizon, the root estimate converges to the finite-horizon value at rate $O(n^{-1/2})$, matching tree-based Stochastic-Power-UCT while reusing samples across shared states. We also study two full-state variants: GS-Power-UCT-F, which stores one node per physical state to increase sample sharing but may mix values from different remaining horizons, and GS-Power-UCT-F$^+$, which uses an adaptive horizon to control this bias. The latter converges to $V^\star(s_0)$, the optimal infinite-horizon discounted value at the root state s_0, when the remaining cross-depth gap vanishes. Experiments on stochastic planning benchmarks show improved sample efficiency over tree-based and graph-based baselines.

1 Introduction

Monte-Carlo Tree Search (MCTS) [3, 2] is a widely used family of online planning algorithms that combine Monte-Carlo sampling with forward tree search. The success of MCTS lies in adaptive exploration strategies inspired by the multi-armed bandit (MAB) literature, most notably the Upper Confidence Bound for Trees (UCT) algorithm [7]. Recent advances in coupling MCTS with deep learning [12, 13, 9] have enabled breakthroughs in complex decision-making problems.

Despite these successes, tree-based MCTS suffers from two fundamental limitations. First, theoretical analysis of the logarithmic exploration bonus in UCT is incomplete due to issues identified by Shah et al. [10], leading to the development of Fixed-Depth-MCTS with polynomial bonuses [11]. Second, and more relevant to this work, tree-based MCTS does not identify states that are reachable via multiple trajectories. When a state s can be reached via two different action sequences, it is represented as two separate nodes in the search tree, leading to redundant exploration and suboptimal use of computational budget.

Leurent and Maillard [8] address the second limitation by proposing Monte-Carlo Graph Search (MCGS), which merges identical states into a single node in a directed graph. They show that this merging can reduce the effective

branching factor from κ (tree) to $\kappa_\infty \leq \kappa$ (graph), yielding improved regret bounds. However, their theoretical analysis is restricted to *deterministic* MDPs with an Optimism in the Face of Uncertainty (OFU) planning style, and the extension to stochastic MDPs is purely empirical.

Separately, the Stochastic-Power-UCT algorithm [5] addresses the first limitation by introducing power mean value estimation with polynomial exploration bonuses for stochastic MCTS, establishing $O(n^{-1/2})$ convergence rate for value estimation at the root node.

Contributions. We make the following contributions:

1. We introduce **Graph-Based Stochastic Power-UCT** (GS-Power-UCT), which combines power mean backups, polynomial exploration bonuses, and depth-augmented graph search. By keying nodes as (s, h), the search graph is a DAG by construction for any stochastic MDP, including MDPs with cycles.
2. We prove that GS-Power-UCT preserves the (α, β)-concentration (Definition 1) guarantees of Stochastic-Power-UCT on depth-augmented graphs. The key step is a generalized graph Q-concentration lemma, Lemma 1, which handles child estimates whose visits are aggregated from multiple parents. This yields root expected error $O(n^{-1/2})$ for the truncated value $\widetilde{V}(s_0, 0)$; see Theorems 6 and 1.
3. We analyze the cost of merging states across depths. Proposition 3 shows that naive full-state merging can introduce irreducible cross-depth bias. For the practical full-merge variant, we give a controlled-bias bound in Theorem 3, and an adaptive-horizon guarantee for GS-Power-UCT-F$^+$ in Theorem 4, where convergence to $V^\star(s_0)$ holds when the empirical cross-depth gap vanishes.
4. We quantify the sample-sharing effect of depth-augmented graph search. For a fixed collection of simulated trajectories, the graph representation is the quotient of the unrolled tree representation obtained by identifying equal (s, h) pairs. Theorem 2 gives a deterministic sample-sharing identity and shows that, under the same recursive proof recipe, the corresponding same-trajectory proof bounds are not enlarged by aggregation. This is a representation-level comparison, not an algorithm-level dominance claim over an independently run tree MCTS algorithm.

2 Preliminaries

2.1 Markov Decision Process

We consider a discrete-time discounted MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, R, P, \gamma \rangle$, where $\mathcal{S}$ is the state space and $\mathcal{A}_s \subseteq \mathcal{A}$ is the nonempty set of admissible actions at s, with $K \triangleq \max_s |\mathcal{A}_s| < \infty$. A simulator response at (s, a) consists of a successor drawn from $P(\cdot \mid s, a)$ and a reward in $[0, R_{\max}]$; $R(s, a, s')$ denotes its conditional mean given s'. The discount factor is $\gamma \in [0, 1)$. The optimal value function satisfies the Bellman optimality equation:

$$V^\star(s) = \max_{a \in \mathcal{A}_s} Q^\star(s, a), \quad Q^\star(s, a) = \sum_{s'} P(s'|s, a) \left[R(s, a, s') + \gamma V^\star(s') \right]. \tag{1}$$

2.2 MCTS with Power Mean Value Backup

Given a planning horizon H and a playout policy π_0 with value V_0, define inductively $\widetilde{V}(s_H) = V_0(s_H)$ and for $h \leq H - 1$:

$$\widetilde{Q}(s_h, a) = r(s_h, a) + \gamma \sum_{s_{h+1}} P(s_{h+1}|s_h, a) \widetilde{V}(s_{h+1}), \quad \widetilde{V}(s_h) = \max_{a \in \mathcal{A}_{s_h}} \widetilde{Q}(s_h, a), \tag{2}$$

where $r(s_h, a)$ is the expected immediate reward; the truncation error satisfies $|Q^\star(s_0, a) - \widetilde{Q}(s_0, a)| \leq \gamma^H \|V^\star - V_0\|_\infty$. Equivalently, with the Bellman optimality operator $(\mathcal{B}V)(s) \triangleq \max_{a \in \mathcal{A}_s} \sum_{s'} P(s'|s, a)[R(s, a, s') + \gamma V(s')]$, $V^\star$ is the unique fixed point of $\mathcal{B}$ and the depth-h truncated value is $\widetilde{V}(s, h) = (\mathcal{B}^{H-h} V_0)(s)$. After t trajectories, the *power mean value estimate* at an internal node s_h is

$$\widehat{V}_t(s_h) = \left(\sum_{a \in \mathcal{A}_{s_h}} \frac{T_{s_h,a}(t)}{t} \left(\widehat{Q}_{T_{s_h,a}(t)}(s_h, a) \right)^p \right)^{1/p}, \tag{3}$$

where $p \in [1, +\infty)$ and $T_{s_h,a}(t)$ is the number of visits to (s_h, a).

2.3 Concentration Framework

Following Shah et al. [11] and Dam et al. [5], we use the following notion of polynomial concentration.

Definition 1 ((α, β)-concentration)**.** *A sequence of estimators $(\widehat{V}_n)_{n\geq 1}$ concentrates at rate (α, β) towards some limit V if there exists a constant $c > 0$ such that:*

$$\forall n \geq 1,\ \forall \varepsilon > 0: \quad \mathbb{P}\left(|\widehat{V}_n - V| > \varepsilon\right) \leq c\, n^{-\alpha} \varepsilon^{-\beta}. \tag{4}$$

We write $\widehat{V}_n \xrightarrow[n\to\infty]{\alpha,\beta} V$.

3 Graph-Based Stochastic Power-UCT

The efficiency of graph-based planning in MDPs hinges on the mechanism used to merge trajectories. We present GS-Power-UCT, an algorithm designed to balance **estimation accuracy** (preserving value consistency) with **sample efficiency** (maximizing state reuse). We formalize this balance through a state-mapping function $\phi(s, h)$ that defines how physical states s and trajectory depths h are indexed in the search graph $\mathcal{G}_n$, allowing us to transition between a theoretically consistent model and a computationally high-throughput variant.

3.1 Search Graph Construction

We maintain a graph $\mathcal{G}_n = (\mathcal{N}_n, \mathcal{E}_n)$ where each node $v \in \mathcal{N}_n$ represents an aggregate state $v = \phi(s, h)$, and the choice of ϕ determines the topology of the search space. The *depth-augmented mapping* $\phi(s, h) = (s, h)$ distinguishes the same physical state at different depths and is the **canonical form** for finite-horizon MDPs, since the value of a state is inherently non-stationary with respect to the remaining horizon $H - h$; it ensures $\mathcal{G}_n$ is a DAG by construction (Prop. 1), eliminating the bias caused by depth-cutoff truncation, and merges all same-depth transpositions, which are the most frequent in structured MDPs. The *full-state mapping* $\phi(s, h) = s$ instead merges all visits to a physical state regardless of depth, drastically reducing the graph size and maximizing information sharing across stages of the trajectory; this relaxation is particularly effective when the environment is cyclic or when V_0 is a strong estimator, making the horizon-induced bias negligible.

3.2 Analyzing the Merging Trade-off

The transition from depth-augmented to full-state merging involves a fundamental trade-off: the former guarantees convergence to the optimal truncated value, whereas the latter introduces a **cross-depth bias** Δ_{CROSS} that we now quantify. For each physical state s, let $\mathcal{D}(s) = \{h : s \text{ is visited at depth } h \text{ during planning}\}$, with $h_{\min}(s) = \min \mathcal{D}(s)$ and $h_{\max}(s) = \max \mathcal{D}(s)$. The cross-depth gap at s is

$$\delta(s) = \max_{h_1, h_2 \in \mathcal{D}(s)} \left| \widetilde{V}(s, h_1) - \widetilde{V}(s, h_2) \right|. \tag{5}$$

Since $\widetilde{V}(s, h) = (\mathcal{B}^{H-h} V_0)(s)$ and $\mathcal{B}$ is a γ-contraction in $\|\cdot\|_\infty$,

$$\delta(s) \leq \frac{\gamma^{H-h_{\max}(s)} - \gamma^{H-h_{\min}(s)}}{1-\gamma} \|\mathcal{B}V_0 - V_0\|_\infty \leq \frac{(1+\gamma)\left(\gamma^{H-h_{\max}(s)} - \gamma^{H-h_{\min}(s)}\right)}{1-\gamma} \|V^\star - V_0\|_\infty, \tag{6}$$

using $\|\mathcal{B}V_0 - V_0\|_\infty \leq (1+\gamma)\|V^\star - V_0\|_\infty$ for the second form. The global cross-depth gap is $\Delta_{\text{CROSS}} = \max_{s\in\mathcal{G}} \delta(s)$. Thus full-state merging is reliable when the Bellman residual of the playout value is small, when relevant depths are close, or when the remaining horizons $H - h$ are all large; if every physical state is visited at a single depth, then $\delta(s) = 0$.

To resolve this, we use **depth-augmented nodes**: the graph $\mathcal{G}_n = (\mathcal{N}_n, \mathcal{E}_n)$ has nodes $(s, h) \in \mathcal{S} \times \{0, \ldots, H\}$, where s is the physical state and h is the trajectory depth at which the node was created. Formally, $\mathcal{N}_n \subseteq \mathcal{S} \times \{0, \ldots, H\}$ collects the depth-augmented nodes discovered so far, and $\mathcal{E}_n$ contains the observed transitions $((s, h), a, (s', h+1))$ with $(s, h), (s', h+1) \in \mathcal{N}_n$ and $a \in \mathcal{A}_s$. Since every edge strictly increases the depth coordinate, $\mathcal{G}_n$ is a DAG by construction. Crucially, two trajectories reaching the same physical state s at the same depth h *are merged* — they share the node (s, h) and its value estimate $\widehat{V}(s, h)$ — capturing the key benefit of graph-based planning while ensuring each node has a well-defined target value $\widetilde{V}(s_h)$ from (2).

Proposition 1 (DAG by construction)**.** *The search graph $\mathcal{G}_n$ maintained by GS-Power-UCT is a DAG for any MDP, including those with cyclic transitions; every edge connects (s, h) to $(s', h+1)$, strictly increasing the depth coordinate, so no directed cycle can exist.*

Table 1: Conditions for algorithmic constants. $h \in [0, H-1]$.

Algorithmic constants (all rows must hold)
$2 < b_h < \alpha_h; \quad \alpha_h(1 - b_h/\alpha_h) \le b_h.$
$\big(1 \le p \le 2,\ \alpha_h \le \frac{\beta_h}{2}\big)$ OR $\big(p > 2,\ \alpha_h \le \frac{\beta_h}{2},\ 0 < \alpha_h - \frac{\beta_h}{p} < 1\big).$
$\alpha_h = (b_{h+1} - 1)(1 - b_{h+1}/\alpha_{h+1}); \quad \beta_h = b_{h+1} - 1.$

Table 2: Comparison of the algorithm variants. $H(n) = \lceil \log(n)/(2\log(1/\gamma)) \rceil$ and $c(n) \triangleq c(H(n))$. The $V^\star(s_0)$ target and displayed rate for F$^+$ are conditional on the cross-depth conditions in Theorem 4.

Variant	**Node key**	**Max nodes**	**DAG?**	**Bias**	**Convergence target**
Tree-MCTS	path	K^H	trivially	0	$\widetilde{V}(s_0), O(n^{-1/2})$
GS-Power-UCT	(s,h)	$\lvert\mathcal{S}\rvert(H+1)$	by constr.	0	$\widetilde{V}(s_0), O(n^{-1/2})$
GS-Power-UCT-F	s	$\lvert\mathcal{S}\rvert$	sometimes	Δ_{CROSS}	$\widetilde{V}(s_0), O(n^{-1/2}) + O(\Delta_{\text{CROSS}})$
GS-Power-UCT-F$^+$	s	$\lvert\mathcal{S}\rvert$	sometimes	$\to 0$	$V^\star(s_0), O(c(n)n^{-1/2})$

The internal nodes $\mathring{\mathcal{G}}_n$ (fully expanded) and boundary nodes $\partial\mathcal{G}_n$ (not yet expanded) partition $\mathcal{N}_n$. For each $(s,h) \in \mathcal{N}_n$ we maintain **global** visit counts $T_{s,h}(n)$ (total visits to (s,h)), $T_{s,h,a}(n)$ (action a taken at (s,h)), and $T^{s'}_{s,h,a}(n)$ (transitions $((s,h),a) \to (s',h+1)$). These aggregate visits from *all* parents reaching (s,h) — the defining feature of global-count graph-based planning.

Remark 1 (Same-depth merging captures most transpositions). The most common transpositions in planning problems occur at the same depth — different action sequences of equal length leading to the same state (e.g., "left then up" vs. "up then left" in a grid) — and the depth-augmented design merges exactly these. States revisited at different depths (e.g., via a cycle) are kept separate, which is both theoretically necessary (different target values) and practically minor (cross-depth transpositions are rarer in finite-horizon planning).

3.3 The Four Steps of GS-Power-UCT

Each simulation t of GS-Power-UCT proceeds in four steps.

Step 1: Selection with Lookup. Starting from the root $(s_0, 0)$, at each internal node $(s,h) \in \mathring{\mathcal{G}}_n$ select an action via the UCB-style rule

$$a = \arg\max_{a \in \mathcal{A}_s} \left\{ \widehat{Q}_{T_{s,h,a}(t)}(s,h,a) + C \cdot \frac{T_{s,h}(t)^{b_{h+1}/\beta_{h+1}}}{T_{s,h,a}(t)^{\alpha_{h+1}/\beta_{h+1}}} \right\}, \tag{7}$$

where $\{b_h\}$, $\{\alpha_h\}$, $\{\beta_h\}$ satisfy Table 1 and $C > 0$ is an exploration constant. Sample $s' \sim P(\cdot|s,a)$ and perform a **depth-augmented lookup**: if $(s', h+1) \in \mathcal{G}_n$ (a *same-depth transposition*), continue selection from the existing node, adding the edge $((s,h),a) \to (s',h+1)$ to $\mathcal{E}_n$ if new; otherwise proceed to Step 2. Selection terminates at depth H.

Step 2: Expansion. If $(s', h+1) \notin \mathcal{G}_n$, add it to $\partial\mathcal{G}_n$ with $T_{s',h+1} = 0$ and append the edge $((s,h),a) \to (s',h+1)$ to $\mathcal{E}_n$.

Step 3: Evaluation. For a new boundary node $(s', h+1) \in \partial\mathcal{G}_n$, draw one fresh rollout $Z_1(s') \sim \pi_0(s')$ with $\mathbb{E}[Z_1(s')] = V_0(s')$, and set $\widehat{V}_1(s', h+1) = Z_1(s')$ and $T_{s',h+1} = 1$. This is a single initialization sample; internal value estimates subsequently updated by the adaptive search are not treated as i.i.d. samples.

Step 4: Backpropagation (Path-Only). Update statistics only along the trajectory $\{(s_0,0), a_0, \ldots, (s_\ell, \ell)\}$. Writing $T \equiv T_{s_h,h,a_h}(t)$ and $T' \equiv T_{s_h,h}(t)$, and letting r_t be the reward observed on the corresponding transition at simulation t, in reverse order:

$$T \leftarrow T + 1, \quad T' \leftarrow T' + 1, \tag{8}$$

$$\widehat{Q}_T(s_h,h,a_h) \leftarrow \frac{(T-1)\widehat{Q}_{T-1}(s_h,h,a_h) + r_t + \gamma \widehat{V}_{T_{s_{h+1},h+1}}(s_{h+1},h+1)}{T}, \tag{9}$$

$$\widehat{V}_{T'}(s_h,h) \leftarrow \Big(\textstyle\sum_{a \in \mathcal{A}_{s_h}} \frac{T_{s_h,h,a}(t)}{T'} \big(\widehat{Q}_{T_{s_h,h,a}(t)}(s_h,h,a)\big)^p \Big)^{1/p}. \tag{10}$$

After n simulations, the recommended action is $\hat{a}_n = \arg\max_{a \in \mathcal{A}_{s_0}} \widehat{Q}_{T_{s_0,0,a}(n)}(s_0, 0, a)$ and the root value estimate is $\widehat{V}_n(s_0, 0)$.

Remark 2 (Path-only vs. full graph backpropagation). We deliberately restrict backpropagation to the current trajectory: full graph backpropagation would propagate information faster but introduces update-ordering issues and circular dependencies that break the analysis, whereas path-only updates preserve the "one simulation, one trajectory" structure essential for the (α, β)-concentration framework.

4 Theoretical Analysis

The depth-augmented graph $\mathcal{G}_n$ maintained by GS-Power-UCT is a DAG by Proposition 1, without any assumption on the MDP topology. This holds even for MDPs with arbitrary cycles or self-loops. The separate probabilistic and finiteness conditions used by the concentration results are collected in Appendix A; the theoretical analysis proceeds by exploiting the DAG structure together with those conditions.

4.1 Graph Structure and Topological Order

Since $\mathcal{G}_n$ is a DAG with depth as the natural topological coordinate, every node (s, h) has a well-defined depth $h \in \{0, 1, \ldots, H\}$. For each node, define:

- $\textsc{Parents}(s, h) = \{((s', h-1), a) : ((s', h-1), a, (s, h)) \in \mathcal{E}_n\}$: the set of parent-action pairs,
- $\mathcal{N}_h = \{(s, h) \in \mathcal{N}_n\}$: the set of nodes at depth h.

The key structural difference from tree-MCTS is that a node (s, h) with $|\textsc{Parents}(s, h)| > 1$ receives visits from multiple parent nodes. In tree-MCTS, the visit count $T_{s,h}(n)$ is entirely controlled by one parent's bandit strategy. In GS-Power-UCT, $T_{s,h}(n) = \sum_{((s',h-1),a) \in \textsc{Parents}(s,h)} T^s_{s',h-1,a}(n)$, aggregating contributions from all parents.

Each node (s, h) has a well-defined concentration target: the truncated value $\widetilde{V}(s_h)$ defined by the Bellman recursion (2) starting from depth h with remaining horizon $H - h$. This is guaranteed because all visits to (s, h) come from depth h — the depth-augmentation prevents mixing estimates from different effective horizons.

Definition 2 (Transposition ratio)**.** *For depth h, define the* transposition ratio

$$\tau_h \triangleq \frac{|\mathcal{N}_h|}{N_h^{\mathcal{T}}}, \tag{11}$$

where $|\mathcal{N}_h|$ is the number of unique depth-augmented nodes at depth h and $N_h^{\mathcal{T}}$ is the number of nodes at depth h in the unrolled tree $\mathcal{T}(\mathcal{G})$. We have $\tau_h \in (0, 1]$, with $\tau_h = 1$ corresponding to no transpositions (tree) and $\tau_h \ll 1$ indicating heavy same-depth state merging.

4.2 Generalized Topological Concentration Lemma

The following lemma is the key technical innovation for extending the concentration framework from trees to graphs. It generalizes Lemma 1 of Dam et al. [5] to handle the case where child value estimates are updated by visits from multiple parents. For each fixed state–depth–action query, the reward–successor pairs form the fresh local stream in condition (A-Fresh) of Assumption 1; only this local stream is i.i.d., whereas the child estimates are handled through their concentration premise.

Lemma 1 (Q-value concentration in graph)**.** *Consider an internal node s in $\mathcal{G}$ with action $a \in \mathcal{A}_s$, and let $M = |\{s' : P(s'|s, a) > 0\}|$. For each $m \in [M]$, suppose the value estimates $(\widehat{V}_{m,n})_{n \geq 1}$ for successor s_m satisfy $\widehat{V}_{m,n} \xrightarrow[n\to\infty]{\alpha,\beta} V_m$ with respect to the **global** visit count of s_m, with $\widehat{V}_{m,n} \leq L$, in the time-uniform form of local hypothesis (H-Q) in Appendix A. Let X_i be i.i.d. rewards with mean $\mu = r(s, a)$, $S_i \sim P(\cdot|s, a)$ the i.i.d. transitions and $N_m^n = \#\{i \leq n : S_i = s_m\}$ the local transition counts. Let $T_{s_m}^{\text{EXT}}(n)$ be the (non-decreasing, possibly random) visits to s_m from* other *parents, and define the* effective *count $T_{s_m}^{\text{EFF}}(n) = N_m^n + T_{s_m}^{\text{EXT}}(n)$. Then, provided $2\alpha \leq \beta$ and $\beta > 1$, the Q-value estimate*

$$\widehat{Q}_n(s, a) = \frac{1}{n} \sum_{i=1}^{n} X_i + \gamma \sum_{m=1}^{M} \frac{N_m^n}{n} \widehat{V}_{m, T_{s_m}^{\text{EFF}}(n)} \tag{12}$$

satisfies $\widehat{Q}_n(s, a) \xrightarrow[n\to\infty]{\alpha,\beta} \mu + \gamma \sum_{m=1}^{M} p_m V_m = \widetilde{Q}(s, a)$, where $p_m = P(s_m|s, a)$.

Remark 3 (Q-value estimator form). The lemma analyzes the estimator $\widehat{Q}_n = \bar{X}_n + \gamma\langle \hat{p}_n, \hat{\boldsymbol{V}}_n \rangle$ where $\hat{\boldsymbol{V}}_n$ uses the child value estimates at their visit counts at time n. The algorithm's running average instead uses the child value estimate *at the time of each individual visit*. These two forms have the same limit $\widetilde{Q}(s,a)$ and the concentration proof applies to both, because the bound on A_2 controls the deviation $|\widehat{V}_{m,k} - V_m|$ for each fixed visit count k, which holds regardless of whether k is an intermediate or final count. This is the same convention adopted in Dam et al. [5], Shah et al. [11].

4.3 Main Result: Root Node Convergence

We can now state the main convergence result, obtained by induction over the topological order of the DAG. For a depth-augmented node (s,h), we write $\widetilde{V}(s,h)$ and $\widetilde{Q}(s,h,a)$ as shorthand for the depth-h truncated Bellman targets $\widetilde{V}(s_h)$ and $\widetilde{Q}(s_h,a)$ from Equation (2).

Theorem 1 (Convergence rate of expected payoff)**.** *Under Assumption 1, at the root depth-augmented node $(s_0,0)$, with optimal parameter tuning, GS-Power-UCT satisfies*

$$\left|\mathbb{E}[\widehat{V}_n(s_0,0)] - \widetilde{V}(s_0,0)\right| \leq O(n^{-1/2}). \tag{13}$$

4.4 Quantifying the Graph Advantage

While Theorems 6 and 1 show that GS-Power-UCT achieves the same *asymptotic rate* $O(n^{-1/2})$ as tree-based Stochastic-Power-UCT, the graph structure provides a concrete advantage through the concentration constants.

Theorem 2 (Sample sharing and non-worsening proof constants)**.** *Fix a depth-augmented search graph $\mathcal{G}$ and its unrolled tree $\mathcal{T}(\mathcal{G})$. For each depth h, let $\phi_h : \mathcal{T}_h \to \mathcal{N}_h$ be the projection from a tree copy to its depth-augmented graph node, and write $\mathcal{C}(v) = \phi_h^{-1}(v)$. Under the natural coupling using the same simulated trajectories,*

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$$T_v^{\mathcal{G}}(n) = \sum_{u \in \mathcal{C}(v)} T_u^{\mathcal{T}}(n), \qquad T_v^{\mathcal{G}}(n) \geq T_u^{\mathcal{T}}(n) \quad \forall u \in \mathcal{C}(v). \tag{14}$$

The same identity holds for action counts $T_{v,a}^{\mathcal{G}}(n)$ and $T_{u,a}^{\mathcal{T}}(n)$. Consequently,

$$\frac{|\mathcal{N}_h|^{-1}\sum_{v\in\mathcal{N}_h} T_v^{\mathcal{G}}(n)}{|\mathcal{T}_h|^{-1}\sum_{u\in\mathcal{T}_h} T_u^{\mathcal{T}}(n)} = \frac{|\mathcal{T}_h|}{|\mathcal{N}_h|} = \frac{1}{\tau_h}. \tag{15}$$

Moreover, apply the same recursive concentration proof to $\mathcal{G}$ and $\mathcal{T}(\mathcal{G})$, with identical leaf constants and Stochastic-Power-UCT parameters. If the recursive constant maps are monotone, in the sense that larger child sample counts and smaller child concentration constants cannot worsen the parent constant, then the proof-generated constants satisfy

$$c_h^{\mathcal{G}}(v) \leq c_h^{\mathcal{T}}(u), \qquad \forall v \in \mathcal{N}_h,\ \forall u \in \mathcal{C}(v). \tag{16}$$

In particular, $c_0^{\mathcal{G}} \leq c_0^{\mathcal{T}}$ at the root.

4.5 Computational Complexity

Proposition 2 (Complexity per simulation)**.** *Each simulation of GS-Power-UCT has the following complexity:*

- ***Selection + Lookup**: $O(H \cdot (K + T_{\text{HASH}}))$, where T_{HASH} is the cost of a hash table lookup. For discrete states with a hash table, $T_{\text{HASH}} = O(1)$ amortized.*
- ***Expansion**: $O(K + T_{\text{HASH}})$ for creating a new node and adding edges.*
- ***Evaluation**: $O(H)$ for a rollout of length H.*
- ***Backpropagation**: $O(H \cdot K)$ for updating Q-values and power means along the path.*

Total per simulation: $O(H \cdot (K + T_{\text{HASH}}))$, which is the same as tree-MCTS up to the T_{HASH} factor.

4.6 Convergence of the Full-Merge Variant

We now analyze the convergence of GS-Power-UCT-F (Algorithm 2), the full state merging variant from Section C.2. The key difference is that a shared node s visited at multiple depths targets a mixture of truncated values rather than a single well-defined limit. We show that the convergence rate is preserved up to an additive bias term.

Theorem 3 (Convergence rate of GS-Power-UCT-F)**.** *Apply GS-Power-UCT-F (Algorithm 2) with algorithmic constants $\{b_h\}_{h=0}^{H-1}$, $\{\alpha_h\}_{h=0}^{H-1}$, $\{\beta_h\}_{h=0}^{H-1}$ satisfying Table 1. If the empirical full-merge target $\bar{V}_n$ from Definition 4 satisfies $\mathbb{E}[|\bar{V}_n(s_0) - \widetilde{V}(s_0,0)|] \le C_\Delta \Delta_{\text{CROSS}}$, then under the optimal tuning $\alpha_0/\beta_0 = 1/2$ the root estimate satisfies*

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0,0)\right| \le \underbrace{O(n^{-1/2})}_{\textit{sampling error}} + \underbrace{O(\Delta_{\text{CROSS}})}_{\textit{cross-depth bias}}. \tag{17}$$

The first term matches tree-based Stochastic-Power-UCT (Theorem 1); the second term vanishes when $\Delta_{\text{CROSS}} = 0$, recovering that rate exactly.

Remark 4 (Practical implications)*.* Theorem 3 suggests a principled choice between variants: use depth-augmented GS-Power-UCT when the cross-depth bias Δ_{CROSS} is comparable to or larger than the sampling error $O(n^{-1/2})$, and GS-Power-UCT-F when $\Delta_{\text{CROSS}} \ll n^{-1/2}$ (good evaluation function, large H, or few cross-depth transpositions) for the smaller graph and greater state sharing.

4.7 Controlling Cross-Depth Bias with Depth-Independent Exploration

The cross-depth bias in Theorem 3 originates from the finite-horizon cutoff: the same physical state s may be entered at different trajectory depths, and hence may be backed up with different remaining horizons. Depth-dependent exploration parameters can introduce an additional artificial source of depth dependence. We remove this artificial source by using a depth-independent exploration bonus. However, this does *not* make the finite-horizon targets $\widetilde{V}_H(s,h)$ independent of h. Therefore, for the full-merge variant we do not claim exact zero cross-depth bias at every local node. Instead, we obtain a root-level convergence bound with an explicit cross-depth bias term, and this term vanishes under a natural depth-slack condition.

Depth-independent exploration bonus. Remark 2 of Dam et al. [5] observes that the optimal parameter choice $\alpha_i/\beta_i = 1/2$ and $b_i/\beta_i = 1/4$ for all $i \in [0, H-1]$ yields the exploration bonus:

$$B(n,s,a) = C \cdot \frac{T_s(n)^{1/4}}{T_{s,a}(n)^{1/2}}, \tag{18}$$

which is *independent of the depth* h. With this choice, the action selection rule at any node s depends only on the global visit counts $(T_s, T_{s,a})$ and Q-value estimates $\widehat{Q}(s,a)$, none of which carry depth information. Consequently, the exploration rule itself no longer depends on the trajectory depth at which s is entered. The remaining source of cross-depth discrepancy is only the finite-horizon cutoff.

If we further choose the horizon H large enough, the cutoff becomes irrelevant: both visits explore the subgraph below s to sufficient depth that the truncation error is negligible. This motivates the *adaptive horizon* strategy.

Adaptive horizon. Given a budget of n simulations, set the planning horizon to

$$H(n) = \left\lceil \frac{\log n}{2\log(1/\gamma)} \right\rceil, \tag{19}$$

so that $\gamma^{H(n)} \le n^{-1/2}$. This choice makes the ordinary root-level truncation error match the $n^{-1/2}$ sampling scale. It does not, by itself, imply that every full-merged node has zero cross-depth bias. To state the required condition precisely, define the adaptive cross-depth gap below.

Definition 3 (Adaptive empirical cross-depth gap)**.** *Fix a horizon H, let $\widetilde{V}_H(s,h)$ be the depth-h truncated Bellman target with terminal value V_0 at depth H from* (2)*, and let $\mathcal{D}_{H,n}(s) \subseteq \{0,\dots,H\}$ be the set of trajectory depths at which physical state s is visited during the first n simulations of the full-merge algorithm with horizon H. The empirical cross-depth gap at horizon H is*

$$\Delta_{\text{CROSS}}^{H,n} = \max_{s:\, \mathcal{D}_{H,n}(s) \neq \emptyset} \; \max_{h_1,h_2 \in \mathcal{D}_{H,n}(s)} \left|\widetilde{V}_H(s,h_1) - \widetilde{V}_H(s,h_2)\right|, \tag{20}$$

and we write $\Delta_{\text{CROSS}}^{n} \triangleq \Delta_{\text{CROSS}}^{H(n),n}$ when $H = H(n) = \lceil \log n/(2\log(1/\gamma))\rceil$.

Theorem 4 (Convergence of GS-Power-UCT-F$^+$ with adaptive horizon)**.** *Consider GS-Power-UCT-F$^+$ with node key s, depth-independent bonus $B(n,s,a) = C\,T_s(n)^{1/4}/T_{s,a}(n)^{1/2}$, and adaptive horizon $H_n \triangleq H(n) = \lceil \log n/(2\log(1/\gamma))\rceil$. Let $\widehat{V}_n^{H_n}(s_0)$ be the root estimate after n simulations at horizon H_n, and assume the hypotheses of Theorem 3 hold for each fixed horizon H with sampling constant $c(H)$ and bias constant C_Δ. Then for every $n \geq 2$,*

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)\right| \leq \underbrace{c(H_n)\,n^{-1/2}}_{\textit{sampling}} + \underbrace{C_\Delta\,\mathbb{E}[\Delta^n_{\text{CROSS}}]}_{\textit{cross-depth bias}} + \underbrace{\gamma^{H_n}\|V^\star - V_0\|_\infty}_{\textit{truncation}}. \tag{21}$$

Since $\gamma^{H_n} \leq n^{-1/2}$, this simplifies to $|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)| \leq (c(H_n) + \|V^\star - V_0\|_\infty)/\sqrt{n} + C_\Delta\,\mathbb{E}[\Delta^n_{\text{CROSS}}]$. Consequently, GS-Power-UCT-F$^+$ is consistent at the root ($\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] \to V^\star(s_0)$) whenever both $c(H_n)\,n^{-1/2} \to 0$ and $\mathbb{E}[\Delta^n_{\text{CROSS}}] \to 0$; if in addition $\mathbb{E}[\Delta^n_{\text{CROSS}}] = O(n^{-1/2})$ and $c(H_n)$ grows at most polylogarithmically in n, the rate is $O(c(H_n)/\sqrt{n})$.

Remark 5 (Scope of Theorem 4). The depth-independent bonus removes depth dependence from the exploration rule, but not from the finite-horizon targets $\widetilde{V}_H(s,h)$. Thus Theorem 4 gives a controlled-bias guarantee through Δ^n_{CROSS}, and consistency follows when $\mathbb{E}[\Delta^n_{\text{CROSS}}] \to 0$.

Remark 6 (On the constant $c(n)$). We write $c(n) = c(H(n))$ for the fixed-horizon concentration constant inherited from Theorem 3. The adaptive result requires $c(H(n))n^{-1/2} \to 0$; in particular, if $c(H)$ grows polynomially in H, then $c(H(n))$ is polylogarithmic in n.

Remark 7 (Connection to depth augmentation). Depth-augmented GS-Power-UCT has zero cross-depth bias for every fixed horizon because nodes are keyed by (s,h). Full-state GS-Power-UCT-F$^+$ uses the smaller key s and can converge to $V^\star(s_0)$ with adaptive $H(n)$, provided the empirical cross-depth gap in Theorem 4 vanishes.

5 Experiments

Planning setting and scope. We study online model-based replanning: at each decision, a planner receives a fixed simulation budget and calls a generative model of the current MDP. No persistent value function is trained across episodes. TD(λ), n-step TD, RTDP, and Dyna instead learn a value function and/or model across interaction, so their learning resource is not directly comparable to a per-decision simulation budget. They are therefore outside this empirical comparison; a learned value function can nevertheless be used as the terminal evaluator V_0.

We evaluate our two graph-based variants on stochastic MDP environments and compare them against four baselines. **UCT** [7] is the classical tree-based MCTS algorithm with a logarithmic exploration bonus and arithmetic-mean backup. **MENTS** [15] incorporates a maximum entropy framework with a softmax backup to facilitate entropy-regularized exploration. **Stochastic-Power-UCT** [5] (P-UCT in the figure and tables) extends the polynomial-bonus design with power mean value backup, establishing $O(n^{-1/2})$ concentration for stochastic MDPs while remaining tree-based. **GBOP** [8] is the graph-based baseline with an OFU planning style. Our **GS-Power-UCT** (Algorithm 1) combines the polynomial bonus and power mean backup with a depth-augmented graph that merges same-depth transpositions, while **GS-Power-UCT-F$^+$** relaxes the merging rule to share a single node per physical state across depths. For F$^+$, at every reported budget n we set $H = H(n)$ once before the search and keep it fixed for all n simulations.

Results. Figure 1 shows that depth-augmented GS-Power-UCT is particularly effective in these stochastic domains at small and moderate simulation budgets, where same-depth transpositions can reuse observations before the budget is exhausted. GS-Power-UCT-F$^+$ is competitive in four of five domains but underperforms in *Passenger Grid*, consistent with its additional cross-depth-bias term. Thus full merging is most appropriate when the cross-depth gap is small (for example, with a strong V_0 or nearby relevant depths); the depth-augmented variant is the robust choice when this is uncertain. The $1/\tau_h$ factor in Theorem 2 is a representation-level sample-sharing identity under the same trajectories, not a predicted performance multiplier for independently run planners.

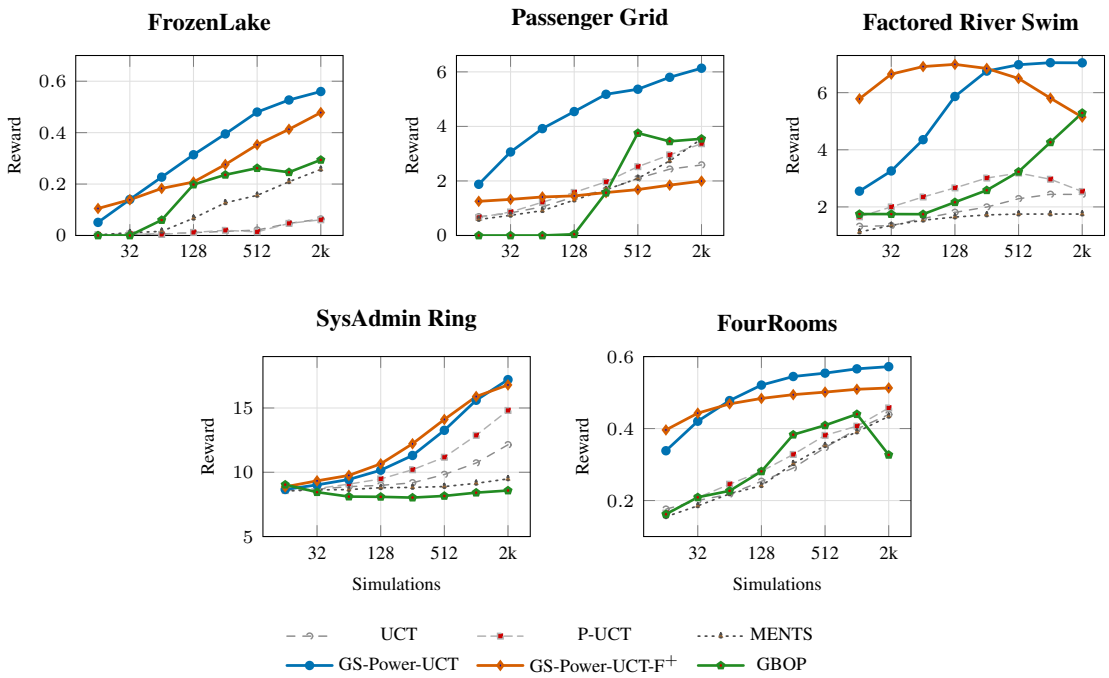


Figure 1: Planning performance versus simulation budget across five stochastic MDPs. Graph-based methods are emphasized with solid curves; tree-based baselines are shown with lighter dashed/dotted curves.

6 Limitations

While **GS-Power-UCT** demonstrates robust theoretical properties, its current formulation involves several caveats that merit further discussion.

- **Non-negative Reward Constraint:** The power-mean backup assumes rewards $R \in [0, R_{\max}]$. Applying GS-Power-UCT to environments with signed rewards would require affine shifts or modified backup operators to ensure value targets remain well-defined.
- **State sharing helps or hurts:** Sharing is useful when multiple simulated trajectories reach the same physical state at the same depth: depth-augmented GS-Power-UCT then pools their samples without mixing finite-horizon targets. The identity in Theorem 2 is representation-level—it does not guarantee that an independently run graph search dominates an independently run tree search. With few or no same-depth transpositions, sharing provides little reuse while retaining hash-table and graph-bookkeeping overhead. Full merging can further reduce the graph size, but it can hurt by mixing targets with different remaining horizons; its error includes the cross-depth term
- **Computational and Memory Overhead:** Maintaining global statistics and hash-based lookups introduces additional overhead. Under depth augmentation, memory usage can scale up to $|\mathcal{S}| \cdot H$ nodes, which may increase the wall-clock cost in large-scale settings despite improved sample efficiency. A detailed empirical breakdown of computational and memory overhead is presented in Appendix E.5.

7 Related Work

MCTS with theoretical guarantees. The theoretical foundations of UCT [7] were revisited by Shah et al. [10, 11], who identified a gap in the analysis of logarithmic exploration bonuses and established polynomial convergence guarantees for Fixed-Depth-MCTS using polynomial bonuses. Complementary work developed a unified analysis of value backup and exploration through power means and convex regularization [24]. For stochastic MDPs, Dam et al. [5] established convergence guarantees for power-mean value estimation, with subsequent work extending theoretical guarantees to continuous stochastic planning [25] and robust planning under reward and transition uncertainty [26]. We extend this line with the first convergence guarantees for graph-based MCTS in stochastic settings.

Online planning and reinforcement learning. TD(λ), n-step TD, Dyna, and RTDP combine learning from interaction with value or model updates [23, 22, 16]. They are complementary to the present work: GS-Power-UCT allocates a fresh online planning budget from a generative model, while a learned value function from such methods can instantiate V_0.

Generative-model planning. SmoothCruiser [27] exploits entropy-regularized Bellman smoothness to obtain sample-complexity guarantees, while SecondOrderSmoothCruiser [28] develops this approach through optimal-transport Bellman smoothing and second-order value estimation. Sparse sampling [20], OLOP [17], and TrailBlazer [19] study planning with a simulator, including methods that address large or even infinite successor spaces. Our concentration proof requires finite successor support and focuses on sharing samples across identical state-depth pairs in the search graph.

Graph-based planning. State merging in planning has appeared in various forms: state aggregation by partitioning [6], MCGS with theoretical analysis for deterministic MDPs [8], belief-state merging in POMDPs without theoretical analysis [1], and transposition-aware/UCD search in game DAGs [18, 21]. Ours provides the first theoretical convergence analysis for graph-based MCTS with UCB-style exploration in stochastic MDPs.

Power mean estimation. The power mean was introduced for MCTS by Dam et al. [4] and later analyzed in Dam et al. [5]; closely related are MENTS [15] and its Rényi/Tsallis variants RENTS and TENTS. Our work is the first to combine power mean estimation with graph-based planning.

8 Conclusion

We introduced Graph-Based Stochastic Power-UCT, the first MCTS algorithm with convergence guarantees that combines graph-based state merging with power mean value estimation for stochastic MDPs. Our key technical contribution is the graph Q-concentration analysis, which allows child estimates to use visits aggregated from multiple same-depth parents. This yields the same $O(n^{-1/2})$ fixed-horizon root rate as tree-based Stochastic-Power-UCT. At the representation level, unrolling the same graph trajectories into a tree shows an explicit sample-sharing advantage: merged graph nodes receive the union of the samples assigned to their tree copies. This can reduce the proof-bound terms associated with transposed states, although it should not be interpreted as a dominance guarantee over an independently run tree-search algorithm.

Several directions for future work are promising. First, connecting the graph-dependent improvement to *spectral properties* of the state graph (effective resistance, spectral gap) would provide more interpretable complexity bounds. Second, extending the analysis to *approximate* state merging for continuous state spaces would broaden the applicability. Third, combining graph-based planning with learned value functions (as in AlphaZero) could yield practical improvements for large-scale problems. Finally, extending the framework to *adversarial* settings (e.g., two-player games with graph transpositions) is a natural next step.

References

[1] J. Ballesteros, L. Merino, M. A. Trujillo, A. Viguria, and A. Ollero. Improving the efficiency of online POMDPs by using belief similarity measures. In *Proc. of ICRA*, 2013.

[2] C. Browne, E. Powley, D. Whitehouse, et al. A survey of Monte Carlo tree search methods. *IEEE Trans. Comput. Intell. AI Games*, 4(1):1–43, 2012.

[3] R. Coulom. Efficient selectivity and backup operators in Monte-Carlo tree search. In *Int. Conf. Computers and Games*, pp. 72–83, 2006.

[4] T. Dam, P. Klink, C. D'Eramo, J. Peters, and J. Pajarinen. Generalized mean estimation in Monte-Carlo tree search. *arXiv preprint arXiv:1911.00384*, 2019.

[5] T. Dam, O.-A. Maillard, and E. Kaufmann. Power mean estimation in stochastic Monte-Carlo tree search. *arXiv preprint arXiv:2406.02235*, 2024.

[6] J. Hostetler, A. Fern, and T. Dietterich. State aggregation in Monte Carlo tree search. In *Proc. of AAAI*, 2014.

[7] L. Kocsis and C. Szepesvári. Bandit based Monte-Carlo planning. In *Proc. of ECML*, pp. 282–293, 2006.

[8] E. Leurent and O.-A. Maillard. Monte-Carlo graph search: the value of merging similar states. In *Proc. of ACML*, pp. 577–592, 2020.

[9] J. Schrittwieser, I. Antonoglou, T. Hubert, et al. Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020.

[10] D. Shah, Q. Xie, and Z. Xu. Non-asymptotic analysis of Monte Carlo tree search. In *SIGMETRICS*, pp. 31–32, 2020.

[11] D. Shah, Q. Xie, and Z. Xu. Nonasymptotic analysis of Monte Carlo tree search. *Operations Research*, 70(6):3234–3260, 2022.

[12] D. Silver, A. Huang, C. J. Maddison, et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016.

[13] D. Silver, J. Schrittwieser, K. Simonyan, et al. Mastering the game of Go without human knowledge. *Nature*, 550(7676):354–359, 2017.

[14] T. Weissman, E. Ordentlich, G. Seroussi, S. Verdu, and M. J. Weinberger. Inequalities for the L_1 deviation of the empirical distribution. *Hewlett-Packard Labs, Tech. Rep.*, 2003.

[15] C. Xiao, R. Huang, J. Mei, D. Schuurmans, and M. Müller. Maximum entropy Monte-Carlo planning. In *NeurIPS*, 2019.

[16] A. G. Barto, S. J. Bradtke, and S. P. Singh. Learning to act using real-time dynamic programming. *Artificial Intelligence*, 72(1–2):81–138, 1995.

[17] S. Bubeck and R. Munos. Open loop optimistic planning. In *Proceedings of the 23rd Conference on Learning Theory*, 2010.

[18] B. E. Childs, J. H. Brodeur, and L. Kocsis. Transpositions and move groups in Monte Carlo tree search. In *Proceedings of the IEEE Symposium on Computational Intelligence and Games*, pp. 389–395, 2008.

[19] J.-B. Grill, M. Valko, and R. Munos. Blazing the trails before beating the path: Sample-efficient Monte-Carlo planning. In *NeurIPS*, 2016.

[20] M. J. Kearns, Y. Mansour, and A. Y. Ng. A sparse sampling algorithm for near-optimal planning in large Markov decision processes. *Machine Learning*, 49(2–3):193–208, 2002.

[21] A. Saffidine, T. Cazenave, and J. Méhat. UCD: Upper confidence bound for rooted directed acyclic graphs. *Knowledge-Based Systems*, 34:26–33, 2011.

[22] R. S. Sutton. Integrated architectures for learning, planning, and reacting based on approximating dynamic programming. In *Proceedings of the Seventh International Conference on Machine Learning*, pp. 216–224, 1990.

[23] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, second edition, 2018.

[24] T. Dam, C. D'Eramo, J. Peters, and J. Pajarinen. A unified perspective on value backup and exploration in Monte-Carlo tree search. *Journal of Artificial Intelligence Research*, 81:511–577, 2024.

[25] T. Q. Dam. Power mean estimation in stochastic continuous Monte-Carlo tree search. In *Proceedings of the 42nd International Conference on Machine Learning*, pp. 12344–12376, 2025.

[26] T. Q. Dam, K. Panaganti, B. Driss, and A. Wierman. Online robust reinforcement learning through Monte-Carlo planning. In *Proceedings of the 42nd International Conference on Machine Learning*, pp. 12314–12343, 2025.

[27] J.-B. Grill, O. D. Domingues, P. Ménard, R. Munos, and M. Valko. Planning in entropy-regularized Markov decision processes and games. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[28] T. Dam. Second-order smooth planning with optimal-transport Bellman smoothing. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026.

Appendix of "Graph-Based Stochastic Power-UCT: Monte-Carlo Graph Search with Power Mean Estimation"

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A Standing Assumptions

This appendix collects the conditions used by the concentration results. They do not restrict the topology of the MDP: cyclic transitions and self-loops remain allowed.

Assumption 1 (Standing MDP and evaluator conditions)**.**

1. ***(A-Finite)*** *$\gamma \in [0,1)$, the planning horizon is finite, every $\mathcal{A}_s$ is finite with $|\mathcal{A}_s| \le K$, and every $P(\cdot \mid s,a)$ has finite support. Thus a (possibly countable) state space has a finite H-reachable graph. Rewards lie in $[0, R_{\max}]$; hence all rollout and backed-up values are bounded by $L \triangleq R_{\max}/(1-\gamma)$.*
2. ***(A-Fresh)*** *A generative-model call at a fixed state–depth–action query returns a fresh reward–successor pair. Successive pairs at that query are i.i.d. from the stationary conditional law. The reward and successor within one pair need not be independent.*
3. ***(A-Rollout)*** *Independent calls to the terminal evaluator return $Z_j(s) \sim \pi_0(s)$ with $\mathbb{E}[Z_j(s)] = V_0(s)$ and $0 \le Z_j(s) \le L$. A call initializes a new boundary node; it is not an i.i.d. assertion about adaptively updated internal estimates.*

Local proof hypotheses (induction hypotheses). (H-Q) In Lemma 1, the child concentration premise is used in time-uniform form: if an adapted traversal selects a child's global count N, then $\mathbb{P}(|\widehat{V}_N - V| > \varepsilon,\ N \ge r) \le c\, r^{-\alpha} \varepsilon^{-\beta}$ for every $r \ge 1$. This induction hypothesis supplies concentration at a possibly random global child count; it does not declare adaptive child estimates i.i.d. **(H-V)** In Lemma 5, the displayed bias-radius condition compares its effective action targets with the depth-specific targets. Corollary 2 instantiates it using the explicit cross-depth bound. For a fixed successor support of size M, graph-Q constants may depend on 2^M and inverse successor probabilities (for example, $\sum_m p_m^{-\alpha}$); rare successors affect constants, not the stated rate. The following table records where these standing conditions and local hypotheses are used.

Table 3: Assumption-discharge table for the analysis.

Result or proof term	**Condition and role**
Leaf concentration	(A-Rollout) gives bounded i.i.d. terminal-evaluator calls, so Hoeffding applies to those calls.
Graph-Q reward term	(A-Finite) bounds rewards; (A-Fresh) makes the local reward–successor-pair stream i.i.d. at a fixed parent-action query.
Graph-Q transition term A_1	(A-Finite) gives finitely many successor categories, and (A-Fresh) gives the i.i.d. successor stream required by the multinomial deviation bound.
Graph-Q child-value term A_2	(H-Q) applies the child concentration premise at its global effective count. Other-parent visits can increase that count; no i.i.d. claim is made for the adaptive child estimates.
Node and root concentration	The graph-Q lemma, the power-mean induction.
Sample-sharing theorem	This is a deterministic representation identity under the same realised trajectories; it introduces no additional stochastic assumption.
Full merge and F^+	(H-V) is instantiated by the explicit cross-depth bound; the stated bounds also require their empirical-target and cross-depth-gap hypotheses. For F^+, $H = H(n)$ is fixed for all simulations in a search with budget n.

B Notation

Table 4: Notation used throughout the paper.

Symbol	Meaning
MDP, returns, and value functions	
$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, R, P, \gamma \rangle$	Discounted MDP with state space $\mathcal{S}$, action space $\mathcal{A}$, conditional-mean reward R, transition kernel P, and discount $\gamma \in [0, 1)$.
$\mathcal{A}_s, K, R_{\max}, L$	Admissible actions at s, $K = \max_s \lvert\mathcal{A}_s\rvert$, reward bound, and return bound $L = R_{\max}/(1-\gamma)$.
$s_0, s, s', a, h;\ i, j, m, t, n$	Root state, generic/current/successor states, action, trajectory depth, and generic sample, successor, visit, or simulation indices.
$H, H(n), H_n$	Fixed planning horizon; adaptive horizon $H(n) = \lceil \log n/(2\log(1/\gamma)) \rceil$; and shorthand $H_n \triangleq H(n)$. Within an F$^+$ search of budget n, $H(n)$ is fixed.
$R(s,a,s'), r(s,a);\ r, r_t, R_i$	Conditional-mean and expected immediate rewards; sampled reward, reward backed up at simulation t, and the i-th sampled reward in the full-merge proof.
π_0, V_0, Z_j, z	Terminal evaluator/policy, its value, its j-th independent output, and a realized rollout/evaluation sample.
$\mathbb{E}, \mathbb{P}, \lVert\cdot\rVert_\infty, \mathbf{1}$	Expectation, probability, sup norm, and indicator function.
$\mathcal{B}$	Bellman optimality operator: $(\mathcal{B}V)(s) = \max_{a\in\mathcal{A}_s} \sum_{s'} P(s'\lvert s,a)[R(s,a,s') + \gamma V(s')]$.
$V^\star, Q^\star$	Infinite-horizon optimal value and action-value functions.
$\widetilde{V}(s,h), \widetilde{Q}(s,h,a), \widetilde{V}_H(s,h)$	Depth-h truncated Bellman targets with terminal value V_0; $\widetilde{V}_H$ displays the horizon explicitly.
$\widehat{V}_t(s,h), \widehat{Q}_t(s,h,a)$ $\widehat{V}_t(s), \widehat{Q}_t(s,a)$	Empirical estimates after t visits in the depth-augmented and full-merge variants, respectively.
p	Power-mean exponent. Rate theorems use finite $p \in [1, \infty)$; $p = \infty$ is an algorithmic implementation limit.
Search graph, unrolling, and sample sharing	
$\phi(s,h)$	Node-key mapping: $\phi(s,h) = (s,h)$ for depth augmentation and $\phi(s,h) = s$ for full merging.
$\mathcal{G}_n = (\mathcal{N}_n, \mathcal{E}_n),\ \mathcal{N}_h$	Search graph after n simulations, its node and edge sets, and the set of discovered depth-h nodes.
$(s,h), \mathring{\mathcal{G}}_n, \partial\mathcal{G}_n, \mathcal{H}$	Depth-augmented node; internal and boundary node sets; and implementation hash map from a node key to its graph node.
PARENTS(s,h)	Parent-action pairs with an edge into depth-augmented node (s,h).
$\mathcal{T}(\mathcal{G}), \mathcal{T}_h;\ \phi_h : \mathcal{T}_h \to \mathcal{N}_h$	Unrolled tree of $\mathcal{G}$, its depth-h nodes, and projection of a tree copy to its graph node.
$v, u, \mathcal{C}(v)$	Graph node, tree copy, and merged-copy class $\mathcal{C}(v) = \phi_h^{-1}(v)$.
$\tau_h, N_h^{\mathcal{T}}$	Transposition ratio $\tau_h = \lvert\mathcal{N}_h\rvert/N_h^{\mathcal{T}}$, where $N_h^{\mathcal{T}} = \lvert\mathcal{T}_h\rvert$ is the number of unrolled-tree nodes at depth h.
$T_v^{\mathcal{G}}, T_u^{\mathcal{T}}, T_{v,a}^{\mathcal{G}}, T_{u,a}^{\mathcal{T}}, N_h(n)$	Graph/tree node and action visit counts in the same-trajectory comparison, and the common total number of visits reaching depth h.
$c_h^{\mathcal{R}}, c_{Q,h}^{\mathcal{R}}, \Phi_h$	Proof-generated value and action-value concentration constants and their monotone Power-UCT aggregation map, for $\mathcal{R} \in \{\mathcal{G}, \mathcal{T}\}$.
Visit counts, bonuses, and algorithmic parameters	
$T_{s,h}(n), T_{s,h,a}(n)$ $T_{s,h,a}^{s'}(n)$	Depth-augmented node visits, action selections, and observed transitions $((s,h),a) \to (s', h+1)$.
$T_s(n), T_{s,a}(n), T_{s,a}^{s'}(n), T_{s,a,h}(n)$	Full-merge state, state-action, successor-transition, and depth-resolved state-action visit counts.
$B(n,s,a), B_h(n,s,a), C$	Exploration bonus (with depth-indexed form when applicable) and exploration constant. For F$^+$, $B(n,s,a) = C T_s(n)^{1/4}/T_{s,a}(n)^{1/2}$.
$b_h, \alpha_h, \beta_h;\ b, \alpha_+, \beta_+$	Depth-indexed Power-UCT parameters and their local full-merge counterparts.
T_{HASH}	Cost of one hash-table lookup in the runtime bound.
$\hat{a}_n$	Recommended root action after n simulations.

Continued on next page

Table 4: Notation used throughout the paper (continued).

Symbol	Meaning
Graph-Q concentration	
$\widehat{V}_n \xrightarrow[n\to\infty]{\alpha,\beta} V;\ \alpha,\beta,c,\varepsilon$	Polynomial concentration: $\mathbb{P}(\lvert\widehat{V}_n - V\rvert > \varepsilon) \le cn^{-\alpha}\varepsilon^{-\beta}$ for every $n \ge 1$ and $\varepsilon > 0$.
$M, [M], s_m, p_m, \boldsymbol{p}, \widehat{\boldsymbol{p}}_n$	Successor-support size, $[M] = \{1,\dots,M\}$, successor states, $p_m = P(s_m \mid s,a)$, and true/empirical successor-probability vectors.
$X_i, S_i, \mu, \bar{X}_n, \boldsymbol{V}, \widehat{\boldsymbol{V}}_n$	Local reward-successor sample, its mean, empirical reward mean, vector of successor targets, and vector of their current estimates.
$N_m^n, \widehat{p}_{m,n}$	Local successor count $N_m^n = \sum_{i=1}^n \mathbf{1}\{S_i = s_m\}$ and empirical probability $\widehat{p}_{m,n} = N_m^n/n$.
$T_{s_m}^{\text{EXT}}(n), T_{s_m}^{\text{EFF}}(n)$	External and effective child visits; $T_{s_m}^{\text{EFF}}(n) = N_m^n + T_{s_m}^{\text{EXT}}(n)$. Subscripts may be abbreviated to m in proofs.
$A_1, A_2, \mathcal{E}_m, r_m$	Empirical-transition and child-value terms in the graph-Q decomposition; the high-local-count event and its threshold.
B_Q, K_m	Uniform graph-Q error bound and proof constant used to dominate exponential tail terms.
Cross-depth and empirical full-merge targets	
$\mathcal{D}(s), \mathcal{D}_{H,n}(s), h_{\min}(s), h_{\max}(s)$	Visited depths of s, their empirical counterpart for horizon H, and their minimum and maximum.
$\delta(s), \Delta_{\text{CROSS}}$	Local and global cross-depth gaps: $\delta(s) = \max_{h_1,h_2\in\mathcal{D}(s)} \lvert\widetilde{V}(s,h_1) - \widetilde{V}(s,h_2)\rvert$ and $\Delta_{\text{CROSS}} = \max_s \delta(s)$.
$\Delta_{\text{CROSS}}^{H,n}, \Delta_{\text{CROSS}}^{n}, C_\Delta$	Empirical cross-depth gap, its adaptive-horizon form $\Delta_{\text{CROSS}}^{H(n),n}$, and the full-merge bias constant.
$\omega_{n,h}(s,a), B_h V, \mathcal{T}_{F,n}^H$	Empirical depth-mixture weight, terminal-value operator, and empirical full-merge Bellman operator.
$\bar{V}(s), \rho_i$	Limiting two-depth mixture and its limiting depth weights in the irreducible-bias proposition.
$\bar{V}_n, \bar{Q}_n$	Unique fixed point of $\mathcal{T}_{F,n}^H$ and its associated empirical full-merge Q-target.
$B_{F,n}^V(s,h)$	Local full-merge target mismatch $\lvert\bar{V}_n(s) - \widetilde{V}(s,h)\rvert$.
$\delta_{Q,t}(s,a;h), \delta_{Q,n}, B_\Delta, B_{\Delta,n}$	Q-level cross-depth drift, its uniform n-simulation bound, and the corresponding bias radii; $B_\Delta = \gamma\Delta_{\text{CROSS}}/(1-\gamma)$ and $B_{\Delta,n} = \delta_{Q,n}/(1-\gamma)$.
Full-merge proof auxiliaries	
$\mathcal{F}_{i,0} \subset \mathcal{F}_{i,1} \subset \mathcal{F}_{i,2}$	Filtration before the i-th transition draw, after that draw, and after the recursive child call.
R_i, S_i, W_i, Y_i	Reward, successor, recursive child return, and backed-up return $Y_i = R_i + \gamma W_i$ in the robust full-merge proof.
A_i, e_i, b_i, ξ_i	Centered transition-reward term, child-estimation error, its predictable part, and its centered part.
$J_i, \rho, \eta_i, G, n_g(t)$	Local successor-depth occurrence count, $\rho = \alpha/\beta$, integration threshold, number of successor-depth groups, and group count up to time t.
$L_R, L_V, L_A, L_\xi, C_{\text{ch}}, C_\rho, \kappa, \sigma_Q$	Robust-proof range, increment, predictable-bias, averaging, and martingale-deviation constants.
$q_a, \bar{q}_a, v_\star, \bar{v}, B$	Depth-specific and effective action targets, their maximizing values, and the local bias radius in hypothesis (H-V).
$\bar{\alpha}, \bar{\beta}, \alpha_V, \beta_V, C_Q, C_V$	Derived Q- and value-concentration exponents and constants in the robust full-merge lemmas.
$X_n, \eta_n, C_{\text{samp}}$	Root absolute error, tail-integration threshold, and resulting sampling-error constant in the full-merge rate proof.

C Detailed Algorithms

C.1 Graph-Based Stochastic Power-UCT and Implementation Specification

Remark 8 (Implementation conventions). The forced-initialization step avoids division by zero in the UCB-style rule when $T_{s,h,a} = 0$. Equivalently, unvisited actions may be assigned infinite exploration bonus until every action has been sampled once.

Algorithm 1 Graph-Based Stochastic Power-UCT (GS-Power-UCT)

Require: Root state s_0, budget n, horizon H, power p, playout policy π_0, exploration constant C

$\mathcal{G} \leftarrow (\{(s_0,0)\},\emptyset)$; $\mathcal{H}[(s_0,0)] \leftarrow$ node; $\partial\mathcal{G} \leftarrow \{(s_0,0)\}$; $\mathring{\mathcal{G}} \leftarrow \emptyset$

Initialize all statistics at $(s_0,0)$: $T_{s_0,0}=0$, $T_{s_0,0,a}=0$, $\widehat{Q}(s_0,0,a)=0$ for all $a \in \mathcal{A}_{s_0}$

for $t = 1,\dots,n$ **do**

  SIMULATEV$(s_0,0,t)$

end for

return $\hat{a}_n = \arg\max_{a\in\mathcal{A}_{s_0}} \widehat{Q}_{T_{s_0,0,a}}(s_0,0,a)$ and $\widehat{V}_n(s_0,0)$

Procedure SIMULATEV(s,h,t)

if $h=H$ or s is terminal **then**

  $z \leftarrow \pi_0(s)$; $\quad T_{s,h} \leftarrow T_{s,h}+1$; $\quad \widehat{V}_{T_{s,h}}(s,h) \leftarrow \widehat{V}_{T_{s,h}}(s,h) + \frac{z-\widehat{V}_{T_{s,h}}(s,h)}{T_{s,h}}$

  return $\widehat{V}_{T_{s,h}}(s,h)$

end if

if $(s,h) \in \partial\mathcal{G}$ **then**

  Move (s,h) from $\partial\mathcal{G}$ to $\mathring{\mathcal{G}}$

end if

if $\exists a \in \mathcal{A}_s$ with $T_{s,h,a}=0$ **then**

  Choose such an action a

else

  $a \leftarrow \arg\max_{a'\in\mathcal{A}_s} \left\{ \widehat{Q}_{T_{s,h,a'}}(s,h,a') + C\frac{T_{s,h}^{\,b_{h+1}/\beta_{h+1}}}{T_{s,h,a'}^{\,\alpha_{h+1}/\beta_{h+1}}} \right\}$

end if

Sample $s' \sim P(\cdot|s,a)$ and $r \sim R(s,a,s')$; set $v'=(s',h+1)$

if $v' \in \mathcal{H}$ **then**

  Add edge $((s,h),a,v')$ to $\mathcal{E}$ if new {same-depth transposition}

  $V_{\text{child}} \leftarrow$ SIMULATEV$(s',h+1,t)$

else

  Add v' to $\mathcal{G}$ and $\partial\mathcal{G}$; $\mathcal{H}[v'] \leftarrow$ node; add edge $((s,h),a,v')$

  Initialize all statistics at v'; $\quad z \leftarrow Z_1(s') \sim \pi_0(s')$; $\quad T_{s',h+1} \leftarrow 1$; $\quad \widehat{V}_{T_{s',h+1}}(s',h+1) \leftarrow z$

  $V_{\text{child}} \leftarrow \widehat{V}_{T_{s',h+1}}(s',h+1)$

end if

$T_{s,h,a}^{s'} \leftarrow T_{s,h,a}^{s'} + 1$; $\quad T_{s,h,a} \leftarrow T_{s,h,a}+1$

$Y \leftarrow r + \gamma V_{\text{child}}$; $\quad \widehat{Q}_{T_{s,h,a}}(s,h,a) \leftarrow \widehat{Q}_{T_{s,h,a}}(s,h,a) + \frac{Y-\widehat{Q}_{T_{s,h,a}}(s,h,a)}{T_{s,h,a}}$

$T_{s,h} \leftarrow T_{s,h}+1$

$\widehat{V}_{T_{s,h}}(s,h) \leftarrow \left(\sum_{a'\in\mathcal{A}_s} \frac{T_{s,h,a'}}{T_{s,h}} \left(\widehat{Q}_{T_{s,h,a'}}(s,h,a')\right)^p\right)^{1/p}$

return $\widehat{V}_{T_{s,h}}(s,h)$

When the depth-augmented lookup finds an existing node $(s', h+1)$, the algorithm does not create a new node and does not perform a fresh evaluation rollout solely because of the transposition. Instead, it continues selection from the existing node and reuses the global statistics already stored at $(s', h+1)$.

Backpropagation remains path-only: only the nodes and edges traversed in the current simulation are updated. Other parents of a transposed node are not updated during this simulation.

By appropriately choosing the algorithmic constants (as specified in Table 1) such that

$$\frac{\alpha_h}{\beta_h} = \frac{1}{2} \quad \text{and} \quad \frac{b_h}{\beta_h} = \frac{1}{4}, \quad \forall h \in [0, H],$$

the optimal convergence rate can be achieved. This choice leads to the following exploration bonus:

$$B_h(n, s, a) = C \, \frac{n^{1/4}}{T_{s,a}(n)^{1/2}},$$

where C denotes a constant controlling the degree of exploration.

As $p \to \infty$, the power mean converges to the maximum operator. Therefore, in this limit, the backup reduces to

$$\widehat{V}_{T_{s,h}}(s, h) \;\leftarrow\; \max_{a' \in \mathcal{A}_s} \widehat{Q}_{T_{s,h,a'}}(s, h, a').$$

This establishes $p = \infty$ as a legitimate parameter regime of the algorithm, in addition to finite p.

C.2 Practical Variant: Full State Merging

While the depth-augmented design of Section 3 provides clean theoretical guarantees, practical MCTS implementations often prefer *full state merging*: keying nodes by state identity alone, without the depth index. This reduces the graph size from at most $|\mathcal{S}|(H+1)$ nodes to at most $|\mathcal{S}|$ nodes, which can be a significant saving in MDPs where the same state is revisited at many different depths (e.g., grid navigation, board games with reversible moves). In this section, we describe this practical variant and analyze the bias it introduces.

Algorithm modification. The only change from Algorithm 1 is in the hash key: the lookup uses $\textsc{Hash}(s')$ instead of $(s', h+1)$. Each physical state s has a single node in the graph, regardless of the depths at which it is visited. The Q-value and visit count statistics are shared across all depths. We present the modified procedure in Algorithm 2.

Source of bias. Consider a state s that is visited at trajectory depths $h_1 < h_2$ during planning. When the simulation reaches s at depth h_1, it continues for $H - h_1$ more steps before reaching the rollout horizon. When it reaches s at depth h_2, it continues for only $H - h_2$ steps. Both trajectories backpropagate returns into the *same* Q-value estimates $\widehat{Q}(s, a)$. The returns from depth h_1 reflect a longer effective planning horizon than those from depth h_2, so they target different values of $\widetilde{Q}$. The resulting Q-value estimate converges to a visit-weighted mixture of the two targets rather than either one individually.

More precisely, the bias propagates recursively: a child s' of s is also visited at multiple depths ($h_1 + 1$ and $h_2 + 1$), so its value estimate $\widehat{V}(s')$ is itself biased, which in turn biases $\widehat{Q}(s, a)$. The total bias at any node depends on the full subgraph of cross-depth transpositions reachable from that node.

Quantifying the bias. For each state s in the graph, define the set of depths at which s is visited:

$$\mathcal{D}(s) = \{h : s \text{ is visited at depth } h \text{ during planning}\}. \tag{22}$$

The *cross-depth gap* at state s is:

$$\delta(s) = \max_{h_1, h_2 \in \mathcal{D}(s)} |\widetilde{V}(s, h_1) - \widetilde{V}(s, h_2)| \leq \frac{\gamma^{H - h_{\max}(s)} - \gamma^{H - h_{\min}(s)}}{1 - \gamma} \|V^\star - V_0\|_\infty, \tag{23}$$

where $h_{\min}(s) = \min \mathcal{D}(s)$, $h_{\max}(s) = \max \mathcal{D}(s)$. The global cross-depth bias is:

$$\Delta_{\text{CROSS}} = \max_{s \in \mathcal{G}} \delta(s). \tag{24}$$

Algorithm 2 Full-Merge Graph-Based Stochastic Power-UCT (GS-Power-UCT-F)

Require: Root state s_0, budget n, horizon H, power p, playout policy π_0, exploration constant C

$\mathcal{G} \leftarrow (\{s_0\}, \emptyset)$; $\mathcal{H}[s_0] \leftarrow$ node; $\partial\mathcal{G} \leftarrow \{s_0\}$; $\mathring{\mathcal{G}} \leftarrow \emptyset$
Initialize all statistics at s_0: $T_{s_0} = 0$, $T_{s_0,a} = 0$, $\widehat{Q}(s_0, a) = 0$ for all $a \in \mathcal{A}_{s_0}$
for $t = 1, \ldots, n$ **do**
  SIMULATEV$(s_0, 0, t)$
end for
return $\hat{a}_n = \arg\max_{a \in \mathcal{A}_{s_0}} \widehat{Q}_{T_{s_0,a}}(s_0, a)$ and $\widehat{V}_n(s_0)$
Procedure SIMULATEV(s, h, t)
if $h = H$ or s is terminal **then**
  $z \leftarrow \pi_0(s)$; $T_s \leftarrow T_s + 1$; $\widehat{V}_{T_s}(s) \leftarrow \widehat{V}_{T_s}(s) + \frac{z - \widehat{V}_{T_s}(s)}{T_s}$
  return $\widehat{V}_{T_s}(s)$
end if
if $s \in \partial\mathcal{G}$ **then**
  Move s from $\partial\mathcal{G}$ to $\mathring{\mathcal{G}}$
end if
if $\exists a \in \mathcal{A}_s$ with $T_{s,a} = 0$ **then**
  Choose such an action a
else
  $a \leftarrow \arg\max_{a' \in \mathcal{A}_s} \left\{ \widehat{Q}_{T_{s,a'}}(s, a') + C \frac{T_s^{b_{h+1}/\beta_{h+1}}}{T_{s,a'}^{\alpha_{h+1}/\beta_{h+1}}} \right\}$
end if
Sample $s' \sim P(\cdot|s, a)$ and $r \sim R(s, a, s')$; set $v' = s'$
if $v' \in \mathcal{H}$ **then**
  Add edge (s, a, v') to $\mathcal{E}$ if new
  $V_{\text{child}} \leftarrow$ SIMULATEV$(s', h + 1, t)$
else
  Add v' to $\mathcal{G}$ and $\partial\mathcal{G}$; $\mathcal{H}[v'] \leftarrow$ node; add edge (s, a, v')
  Initialize all statistics at v'; $z \leftarrow$ rollout from $\pi_0(s')$; $T_{s'} \leftarrow 1$; $\widehat{V}_{T_{s'}}(s') \leftarrow z$
  $V_{\text{child}} \leftarrow \widehat{V}_{T_{s'}}(s')$
end if
$T_{s,a}^{s'} \leftarrow T_{s,a}^{s'} + 1$; $T_{s,a} \leftarrow T_{s,a} + 1$
$Y \leftarrow r + \gamma V_{\text{child}}$; $\widehat{Q}_{T_{s,a}}(s, a) \leftarrow \widehat{Q}_{T_{s,a}}(s, a) + \frac{Y - \widehat{Q}_{T_{s,a}}(s,a)}{T_{s,a}}$
$T_s \leftarrow T_s + 1$
$\widehat{V}_{T_s}(s) \leftarrow \left(\sum_{a' \in \mathcal{A}_s} \frac{T_{s,a'}}{T_s} \left(\widehat{Q}_{T_{s,a'}}(s, a') \right)^p \right)^{1/p}$
return $\widehat{V}_{T_s}(s)$

When is Δ_{CROSS} small? The cross-depth bias is negligible in several important regimes:

1. **Good evaluation function**: When $V_0 \approx V^\star$ (e.g., a well-trained neural network as in AlphaZero), $\|V^\star - V_0\|_\infty$ is small, making Δ_{CROSS} small regardless of the depth gap.
2. **Large horizon**: When H is large relative to the depth range, both $\gamma^{H - h_{\max}}$ and $\gamma^{H - h_{\min}}$ are small, so their difference is negligible.
3. **Small depth range**: When states are revisited only at similar depths ($h_{\max}(s) - h_{\min}(s)$ is small for all s), the cross-depth gap is small.
4. **No cross-depth transpositions**: When every transposition occurs at the same depth (common in grid worlds with deterministic step costs), $\Delta_{\text{CROSS}} = 0$ and the full-merge variant is exactly equivalent to the depth-augmented variant.

D Detailed Proofs

D.1 Proof of Lemma 1

We provide the complete proof of the Generalized Topological Concentration Lemma.

Proof. The case $\gamma = 0$ reduces to the concentration of the empirical reward average, so assume $\gamma > 0$.

By (A-Fresh) in Assumption 1, $(X_i, S_i)_{i\geq 1}$ is the i.i.d. local simulator stream for the fixed parent-action pair. No independence between X_i and S_i within a pair is used. The estimates $\widehat{V}_{m,T_m^{\text{EFF}}(n)}$ are adaptive; their contribution is controlled by the time-uniform local hypothesis (H-Q), rather than by an i.i.d. argument.

Let

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \widehat{\boldsymbol{V}}_n = \left(\widehat{V}_{1,T_1^{\text{EFF}}(n)}, \ldots, \widehat{V}_{M,T_M^{\text{EFF}}(n)}\right),$$

and let

$$\boldsymbol{V} = (V_1, \ldots, V_M), \qquad \widehat{\boldsymbol{p}}_n = (\widehat{p}_{1,n}, \ldots, \widehat{p}_{M,n}) \qquad \left(\widehat{p}_{i,n} = \frac{N_i^n}{n}\right) \qquad \boldsymbol{p} = (p_1, \ldots, p_M).$$

We have

$$\widehat{Q}_n(s,a) = \bar{X}_n + \gamma\langle \widehat{\boldsymbol{p}}_n, \widehat{\boldsymbol{V}}_n\rangle.$$

Therefore,

$$\begin{aligned}
&\mathbb{P}\Big(\Big|\widehat{Q}_n(s,a) - (\mu + \gamma\langle \boldsymbol{p}, \boldsymbol{V}\rangle)\Big| \geq \varepsilon\Big) \\
&\quad \leq \mathbb{P}\Big(|\bar{X}_n - \mu| \geq \frac{\varepsilon}{2}\Big) + \mathbb{P}\Big(\gamma\Big|\langle \widehat{\boldsymbol{p}}_n, \widehat{\boldsymbol{V}}_n\rangle - \langle \boldsymbol{p}, \boldsymbol{V}\rangle\Big| \geq \frac{\varepsilon}{2}\Big) \\
&\quad \leq \mathbb{P}\Big(|\bar{X}_n - \mu| \geq \frac{\varepsilon}{2}\Big) + A_1 + A_2,
\end{aligned} \tag{25}$$

where

$$A_1 = \mathbb{P}\left(\Big|\langle \widehat{\boldsymbol{p}}_n - \boldsymbol{p}, \widehat{\boldsymbol{V}}_n\rangle\Big| \geq \frac{\varepsilon}{4\gamma}\right),$$

and

$$A_2 = \mathbb{P}\left(\Big|\langle \boldsymbol{p}, \widehat{\boldsymbol{V}}_n - \boldsymbol{V}\rangle\Big| \geq \frac{\varepsilon}{4\gamma}\right).$$

Reward term. Assume the reward samples have bounded range at most $R_{\max}$. By Hoeffding's inequality,

$$\mathbb{P}\left(|\bar{X}_n - \mu| \geq \frac{\varepsilon}{2}\right) \leq 2\exp\left(-\frac{n\varepsilon^2}{2R_{\max}^2}\right). \tag{26}$$

Empirical-transition term. Since $|\widehat{V}_{m,T_m^{\text{EFF}}(n)}| \leq L$ for every m,

$$\Big|\langle \widehat{\boldsymbol{p}}_n - \boldsymbol{p}, \widehat{\boldsymbol{V}}_n\rangle\Big| \leq \|\widehat{\boldsymbol{p}}_n - \boldsymbol{p}\|_1 \|\widehat{\boldsymbol{V}}_n\|_\infty \leq L\|\widehat{\boldsymbol{p}}_n - \boldsymbol{p}\|_1.$$

Using the Weissman–type L_1 deviation inequality [14] for the empirical distribution on M categories,

$$A_1 \leq \mathbb{P}\left(\|\widehat{\boldsymbol{p}}_n - \boldsymbol{p}\|_1 \geq \frac{\varepsilon}{4\gamma L}\right) \leq 2^M \exp\left(-\frac{n\varepsilon^2}{32\gamma^2 L^2}\right). \tag{27}$$

Child-value term. Let

$$Z_m(n) = \widehat{V}_{m,T_m^{\text{EFF}}(n)} - V_m, \qquad \delta = \frac{\varepsilon}{4\gamma}.$$

Then

$$A_2 = \mathbb{P}\left(\left|\sum_{m=1}^{M} p_m Z_m(n)\right| \geq \delta\right) \leq \mathbb{P}\left(\sum_{m=1}^{M} p_m |Z_m(n)| \geq \delta\right).$$

Since $\sum_m p_m = 1$, if $|Z_m(n)| < \delta$ for every m, then $\sum_m p_m |Z_m(n)| < \delta$. Hence

$$A_2 \leq \sum_{m=1}^{M} \mathbb{P}(|Z_m(n)| \geq \delta)\,. \tag{28}$$

For each m, define the high-local-count event

$$\mathcal{E}_m = \left\{ N_m^n > \frac{np_m}{2} \right\}.$$

Then

$$\mathbb{P}(|Z_m(n)| \geq \delta) \leq \mathbb{P}(|Z_m(n)| \geq \delta, \mathcal{E}_m) + \mathbb{P}(\mathcal{E}_m^c). \tag{29}$$

On $\mathcal{E}_m$,

$$T_m^{\text{EFF}}(n) = N_m^n + T_m^{\text{EXT}}(n) \geq N_m^n > \frac{np_m}{2}.$$

Let

$$r_m = \max\left\{1, \frac{np_m}{2}\right\}.$$

By the time-uniform child premise (H-Q), we have

$$\begin{aligned} \mathbb{P}\left(|Z_m(n)| \geq \delta, \mathcal{E}_m\right) &\leq \mathbb{P}\left(|\widehat{V}_{m,T_m^{\text{EFF}}(n)} - V_m| \geq \delta, T_m^{\text{EFF}}(n) \geq r_m\right) \\ &\leq c_V 2^\alpha p_m^{-\alpha} n^{-\alpha} \delta^{-\beta}. \end{aligned} \tag{30}$$

Also, since $N_m^n \sim \text{Binomial}(n, p_m)$, Hoeffding's inequality gives

$$\mathbb{P}(\mathcal{E}_m^c) = \mathbb{P}\left(N_m^n \leq \frac{np_m}{2}\right) \leq \exp\left(-\frac{np_m^2}{2}\right). \tag{31}$$

Combining (28), (30), and (31), and substituting $\delta = \varepsilon/(4\gamma)$, we obtain

$$A_2 \leq c_V 2^\alpha (4\gamma)^\beta \left(\sum_{m=1}^{M} p_m^{-\alpha}\right) n^{-\alpha} \varepsilon^{-\beta} + \sum_{m=1}^{M} \exp\left(-\frac{np_m^2}{2}\right). \tag{32}$$

Dominating the exponential terms. The estimators and rewards are bounded, so there exists a deterministic constant $B_Q < \infty$ such that

$$\left|\widehat{Q}_n(s,a) - \widetilde{Q}(s,a)\right| \leq B_Q \qquad \text{almost surely for all } n.$$

For $\varepsilon > B_Q$, the desired probability is zero, so it remains to consider $0 < \varepsilon \leq B_Q$.

Because $2\alpha \leq \beta$, for every $c_0 > 0$ there exists a finite constant $K(c_0, \alpha, \beta)$ such that

$$\exp(-c_0 n\varepsilon^2) \leq K(c_0, \alpha, \beta) n^{-\alpha} \varepsilon^{-\beta} \qquad \forall n \geq 1,\ \varepsilon > 0.$$

Indeed, with $x = n\varepsilon^2$,

$$\exp(-c_0 x) \leq K x^{-\beta/2} = K n^{-\beta/2} \varepsilon^{-\beta} \leq K n^{-\alpha} \varepsilon^{-\beta},$$

where the last inequality uses $\beta/2 \geq \alpha$.

Similarly, for each m,

$$\exp\left(-\frac{np_m^2}{2}\right) \leq K_m n^{-\alpha} \leq K_m B_Q^\beta n^{-\alpha} \varepsilon^{-\beta} \qquad \text{for } 0 < \varepsilon \leq B_Q.$$

Thus the reward term (26), the empirical-transition term (27), and the binomial lower-tail terms in (32) are all bounded by constants times $n^{-\alpha}\varepsilon^{-\beta}$.

Combining these bounds in (25), there exists a constant $C_Q < \infty$ such that

$$\mathbb{P}\left(\left|\widehat{Q}_n(s,a) - \widetilde{Q}(s,a)\right| > \varepsilon\right) \leq C_Q n^{-\alpha} \varepsilon^{-\beta}.$$

This proves the claimed (α, β)-concentration. □

Before providing the Proof for Theorem 6, we provide the proof below.

Lemma 2 (Leaf concentration)**.** *For any leaf node s at depth H in $\mathcal{G}$, the value estimate $\widehat{V}_n(s)$ (average of n i.i.d. calls to π_0) satisfies:*

$$\widehat{V}_n(s) \xrightarrow[n\to\infty]{\alpha_H,\beta_H} \widetilde{V}(s), \tag{33}$$

where α_H, β_H can be chosen to satisfy $\alpha_H \le \beta_H/2$ with $\beta_H > 2$, and $\widetilde{V}(s) = V_0(s)$ is the playout value.

Proof. Since $\widehat{V}_n(s) = \frac{1}{n}\sum_{i=1}^n X_i$ where $X_i \sim \pi_0(s)$ are i.i.d. with mean $V_0(s)$ and bounded in $[0, R_{\max}/(1-\gamma)]$, Hoeffding's inequality gives $\mathbb{P}(|\widehat{V}_n(s) - V_0(s)| > \varepsilon) \le 2\exp(-2n\varepsilon^2/L^2)$, which implies (α,β)-concentration for any $\alpha \le \beta/2$ with appropriate constants. □

Theorem 5 (Power mean concentration at graph nodes)**.** *Consider an internal node s in $\mathcal{G}$ at graph depth h. Suppose that for all actions $a \in \mathcal{A}_s$, the Q-value estimates satisfy $\widehat{Q}_n(s,h,a) \xrightarrow[n\to\infty]{\alpha_{h+1},\beta_{h+1}} \widetilde{Q}(s,h,a)$. Assume the action selection follows* (7) *with parameters satisfying Table 1.*

If $p, \alpha_{h+1}, \beta_{h+1}, b_{h+1}$ satisfy either:

(i) $1 \le p \le 2$ *and* $\alpha_{h+1} \le \beta_{h+1}/2$*, or*

(ii) $p > 2$ *and* $0 < \alpha_{h+1} - \beta_{h+1}/p < 1$*,*

then the power mean value estimate $\widehat{V}_n(s)$ satisfies:

$$\widehat{V}_n(s,h) \xrightarrow[n\to\infty]{\alpha_h,\beta_h} \widetilde{V}(s,h), \tag{34}$$

where $\alpha_h = (b_{h+1}-1)(1 - b_{h+1}/\alpha_{h+1})$ and $\beta_h = (b_{h+1}-1)$.

Proof. This follows directly from Theorem 1 of Dam et al. [5]. The key observation is that Theorem 1 only requires that the Q-value estimates at the node concentrate at rate $(\alpha_{h+1}, \beta_{h+1})$ — it does not require any assumption about *how* those Q-values were formed (tree or graph). Since Lemma 1 establishes exactly this concentration for graph-based Q-values, the theorem applies. □

D.2 Proofs of Theorems 6 and 1

Before going into the detailed proof, we state this result

Theorem 6 (Convergence of GS-Power-UCT)**.** *Apply GS-Power-UCT (Algorithm 1) to an MDP satisfying Assumption 1, with algorithmic constants $\{b_h\}_{h=0}^H$, $\{\alpha_h\}_{h=0}^H$, and $\{\beta_h\}_{h=0}^H$ satisfying Table 1. Then the depth-augmented search graph $\mathcal{G}_n = (\mathcal{N}_n, \mathcal{E}_n)$ is a DAG by Proposition 1. Moreover, the following concentration statements hold.*

(i) For any depth-augmented node $(s,h) \in \mathcal{N}_n$ with $h \in \{0,1,\dots,H\}$, the value estimate stored at this node concentrates around its depth-h truncated Bellman target:

$$\widehat{V}_n(s,h) \xrightarrow[n\to\infty]{\alpha_h,\beta_h} \widetilde{V}(s,h). \tag{35}$$

(ii) For any internal depth-augmented node $(s,h) \in \mathring{\mathcal{G}}_n$ with $h \in \{0,1,\dots,H-1\}$ and any action $a \in \mathcal{A}_s$, the Q-value estimate stored at this node-action pair concentrates around its depth-h truncated Q-target:

$$\widehat{Q}_n(s,h,a) \xrightarrow[n\to\infty]{\alpha_{h+1},\beta_{h+1}} \widetilde{Q}(s,h,a), \qquad \forall a \in \mathcal{A}_s. \tag{36}$$

Proof. We prove by strong induction on depth h, from H down to 0. The induction is well-defined because the depth-augmented graph is a DAG by Proposition 1: every edge connects depth h to depth $h+1$, so all children of a depth-h node have depth exactly $h+1$.

Base case ($h = H$): By Lemma 2, for any node (s,H) at depth H:

$$\widehat{V}_n(s,H) \xrightarrow[n\to\infty]{\alpha_H,\beta_H} \widetilde{V}(s,H) = V_0(s). \tag{37}$$

Inductive step: Suppose (i) and (ii) hold for all depths $h+1, h+2, \dots, H$. Consider a node (s,h) at depth h.

Step 1: Q-value concentration. For any action $a \in \mathcal{A}_s$, the successor nodes are of the form $(s', h+1)$ with $s' \sim P(\cdot|s,a)$, which have depth exactly $h+1$ by the depth-augmented construction. By the inductive hypothesis:

$$\widehat{V}_n(s', h+1) \xrightarrow[n\to\infty]{\alpha_{h+1},\beta_{h+1}} \widetilde{V}(s', h+1), \quad \text{for all successors } s'. \tag{38}$$

Crucially, this concentration targets the correct value $\widetilde{V}(s', h+1)$ because every visit to node $(s', h+1)$ occurs at depth $h+1$ — the depth-augmentation guarantees no mixing of estimates from different effective horizons.

By Lemma 1:

$$\widehat{Q}_n(s, h, a) \xrightarrow[n\to\infty]{\alpha_{h+1},\beta_{h+1}} \widetilde{Q}(s, h, a), \quad \forall a \in \mathcal{A}_s. \tag{39}$$

This establishes (ii) at depth h.

Step 2: Value concentration via power mean. At node s, the action selection rule (7) and the power mean backup (10) satisfy the conditions of Theorem 5 (which is Theorem 1 of Dam et al. [5]). The Q-value estimates concentrate at rate $(\alpha_{h+1}, \beta_{h+1})$ by Step 1, and the parameters satisfy Table 1. Therefore:

$$\widehat{V}_n(s, h) \xrightarrow[n\to\infty]{\alpha_h,\beta_h} \widetilde{V}(s, h), \tag{40}$$

with $\alpha_h = (b_{h+1} - 1)(1 - b_{h+1}/\alpha_{h+1})$ and $\beta_h = (b_{h+1} - 1)$.

This establishes (i) at depth h, completing the induction. □

Proof. Identical to Theorem 3 of Dam et al. [5]. Using Jensen's inequality and the (α_0, β_0)-concentration from Theorem 6:

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0)\right| \le \mathbb{E}\left[|\widehat{V}_n(s_0) - \widetilde{V}(s_0)|\right] = \int_0^\infty \mathbb{P}(|\widehat{V}_n(s_0) - \widetilde{V}(s_0)| \ge \varepsilon)\, d\varepsilon \tag{41}$$

$$\le n^{-\alpha_0/\beta_0} + \frac{c_0}{\beta_0 - 1} n^{-\alpha_0/\beta_0}. \tag{42}$$

Since $\alpha_0/\beta_0 \le 1/2$ (from Table 1), the best achievable rate is $O(n^{-1/2})$, attained by choosing $\alpha_h/\beta_h = 1/2$ for all h. □

D.3 Proof of Theorem 2

Proof. Fix a depth-augmented search graph $\mathcal{G}$ and let $\mathcal{T}(\mathcal{G})$ be its unrolled tree. For each depth h, the projection

$$\phi_h : \mathcal{T}_h \to \mathcal{N}_h$$

maps a tree copy to the unique depth-augmented graph node with the same physical state and the same depth. For a graph node $v \in \mathcal{N}_h$, write

$$\mathcal{C}(v) \triangleq \phi_h^{-1}(v)$$

for the set of tree copies that are merged into v.

The proof has two parts. First, we prove the deterministic sample-sharing identities. Second, we prove the non-worsening comparison for the proof-generated concentration constants.

Coupling the graph and the unrolled tree. We use the natural coupling in which the graph algorithm and the unrolled tree are exposed to the same realized simulated trajectories. The unrolled tree keeps two prefixes distinct whenever their histories differ, even if they end at the same state-depth pair. The graph identifies all prefixes ending at the same depth-augmented node (s, h).

Fix a depth h and a graph node $v \in \mathcal{N}_h$. For the i-th simulated trajectory, let $U_{i,h}$ denote the depth-h prefix in the unrolled tree, whenever the trajectory reaches depth h. Let

$$V_{i,h} = \phi_h(U_{i,h})$$

be its graph projection. Then the event that the graph trajectory visits v is exactly the disjoint union of the events that the unrolled trajectory visits one of the tree copies in $\mathcal{C}(v)$. Hence, at the level of indicators,

$$\mathbf{1}\{V_{i,h} = v\} = \sum_{u \in \mathcal{C}(v)} \mathbf{1}\{U_{i,h} = u\}.$$

Summing over the first n simulations gives

$$T_v^{\mathcal{G}}(n) = \sum_{u \in \mathcal{C}(v)} T_u^{\mathcal{T}}(n). \tag{43}$$

Because every term in the sum is nonnegative, we immediately obtain

$$T_v^{\mathcal{G}}(n) \geq T_u^{\mathcal{T}}(n), \qquad \forall u \in \mathcal{C}(v). \tag{44}$$

The same argument applies after conditioning on the selected action. Indeed, for any action a,

$$\mathbf{1}\{V_{i,h} = v,\ a_{i,h} = a\} = \sum_{u \in \mathcal{C}(v)} \mathbf{1}\{U_{i,h} = u,\ a_{i,h} = a\}.$$

Therefore,

$$T_{v,a}^{\mathcal{G}}(n) = \sum_{u \in \mathcal{C}(v)} T_{u,a}^{\mathcal{T}}(n), \tag{45}$$

and hence

$$T_{v,a}^{\mathcal{G}}(n) \geq T_{u,a}^{\mathcal{T}}(n), \qquad \forall u \in \mathcal{C}(v),\ \forall a \in \mathcal{A}. \tag{46}$$

Average sample-sharing factor. Every simulated trajectory that reaches depth h contributes exactly one depth-h visit in the graph representation and exactly one depth-h visit in the unrolled tree representation. Therefore,

$$\sum_{v \in \mathcal{N}_h} T_v^{\mathcal{G}}(n) = \sum_{u \in \mathcal{T}_h} T_u^{\mathcal{T}}(n). \tag{47}$$

Let this common total be denoted by $N_h(n)$. If $N_h(n) = 0$, then no depth-h node is visited and the identity is vacuous. Otherwise,

$$\frac{1}{|\mathcal{N}_h|} \sum_{v \in \mathcal{N}_h} T_v^{\mathcal{G}}(n) = \frac{N_h(n)}{|\mathcal{N}_h|}, \qquad \frac{1}{|\mathcal{T}_h|} \sum_{u \in \mathcal{T}_h} T_u^{\mathcal{T}}(n) = \frac{N_h(n)}{|\mathcal{T}_h|}.$$

Dividing the two displays gives

$$\frac{|\mathcal{N}_h|^{-1} \sum_{v \in \mathcal{N}_h} T_v^{\mathcal{G}}(n)}{|\mathcal{T}_h|^{-1} \sum_{u \in \mathcal{T}_h} T_u^{\mathcal{T}}(n)} = \frac{|\mathcal{T}_h|}{|\mathcal{N}_h|} = \frac{1}{\tau_h}. \tag{48}$$

Thus $1/\tau_h$ is an average sample-sharing factor over visited depth-h nodes.

Concentration constants at leaves. We now prove the proof-bound comparison by backward induction on depth. At depth H, the target of both a graph leaf $v = (s, H)$ and every tree copy $u \in \mathcal{C}(v)$ is the same terminal value,

$$\widetilde{V}(s, H) = V_0(s).$$

The leaf estimator is an empirical average of rollout or evaluation samples. The leaf concentration constant produced by the same proof recipe is therefore the same for the graph node and for each tree copy. Denote this common leaf constant by c_H^{leaf}. Thus

$$c_H^{\mathcal{G}}(v) = c_H^{\mathrm{leaf}} = c_H^{\mathcal{T}}(u), \qquad \forall u \in \mathcal{C}(v).$$

Moreover, since $T_v^{\mathcal{G}}(n) \geq T_u^{\mathcal{T}}(n)$ by (44), the full leaf concentration bound is no worse in the graph:

$$c_H^{\mathcal{G}}(v)\big(T_v^{\mathcal{G}}(n)\big)^{-\alpha_H} \varepsilon^{-\beta_H} \leq c_H^{\mathcal{T}}(u)\big(T_u^{\mathcal{T}}(n)\big)^{-\alpha_H} \varepsilon^{-\beta_H}.$$

Inductive hypothesis. Assume that the constant comparison holds at depth $h+1$. That is, for every graph child $w \in \mathcal{G}_{h+1}$ and every tree copy $z \in \phi_{h+1}^{-1}(w)$,

$$c_{h+1}^{\mathcal{G}}(w) \leq c_{h+1}^{\mathcal{T}}(z). \tag{49}$$

We show that the same comparison holds at depth h.

Fix a graph node

$$v = (s, h) \in \mathcal{N}_h$$

and a tree copy

$$u \in \mathcal{C}(v) = \phi_h^{-1}(v).$$

The graph node and the tree copy have the same physical state s and the same depth h, so they have the same finite-horizon Bellman target

$$\widetilde{V}(s, h).$$

This is the key reason the depth-augmented representation is analytically clean: same-depth tree copies can be merged without mixing different remaining horizons.

Action-level Q comparison. Fix an action $a \in \mathcal{A}_s$. Let

$$s_1, \dots, s_M$$

be the possible successor states under $P(\cdot|s,a)$, and write

$$w_m = (s_m, h+1)$$

for the corresponding graph child. For the selected tree copy u, let z_m be the tree child reached after action a and successor state s_m. Then

$$z_m \in \phi_{h+1}^{-1}(w_m).$$

Consider the first n visits to the parent-action pair represented by the tree copy (u,a). Under the natural coupling, these same local transition samples also appear among the visits to the graph parent-action pair (v,a), because the graph aggregates all copies of v. Therefore, for each successor s_m, the graph child receives at least the local visits received by the corresponding tree child. More explicitly, if N_m^n denotes the local number of observed transitions to s_m through the selected tree copy, then the graph effective child count has the form

$$T_m^{\text{EFF},\mathcal{G}}(n) = N_m^n + T_m^{\text{EXT},\mathcal{G}}(n),$$

where $T_m^{\text{EXT},\mathcal{G}}(n)$ counts visits to the same graph child from other parent copies. Hence

$$T_m^{\text{EFF},\mathcal{G}}(n) \geq N_m^n = T_m^{\text{EFF},\mathcal{T}}(n). \tag{50}$$

The Q-concentration proof decomposes the estimation error into three types of terms:

$$\text{reward error} \;+\; \text{empirical-transition error} \;+\; \text{child-value error}.$$

For the reward term, the graph proof uses the same reward samples as the tree copy, plus possibly additional samples from other copies. Since the reward concentration bound is nonincreasing in the number of samples, this term cannot be worse in the graph proof.

For the empirical-transition term, the graph again has at least the local transition samples of the selected tree copy. The empirical distribution concentration term is therefore no worse in the graph proof.

For the child-value term, the induction hypothesis gives

$$c_{h+1}^{\mathcal{G}}(w_m) \leq c_{h+1}^{\mathcal{T}}(z_m), \qquad m = 1, \dots, M.$$

In addition, the effective child sample count in the graph is no smaller than the corresponding tree-copy child count by (50). Since the recursive Q-propagation constant is assumed to be coordinatewise nondecreasing in the child value-concentration constants and coordinatewise nonincreasing in the effective child sample counts, the graph action-level Q constant cannot exceed the tree-copy action-level Q constant:

$$c_{Q,h}^{\mathcal{G}}(v,a) \leq c_{Q,h}^{\mathcal{T}}(u,a). \tag{51}$$

This holds for every action $a \in \mathcal{A}_s$.

Value-constant comparison. The Power-UCT value-concentration theorem maps the vector of action-level Q-concentration constants into a value-concentration constant. Write this map abstractly as

$$c_h^{\mathcal{R}}(x) = \Phi_h\left(\{c_{Q,h}^{\mathcal{R}}(x,a)\}_{a\in\mathcal{A}_s}\right), \qquad \mathcal{R} \in \{\mathcal{G}, \mathcal{T}\}.$$

The same Power-UCT parameters are used in both representations, so the map Φ_h is the same for the graph node and the tree copy. By assumption, Φ_h is coordinatewise nondecreasing. Combining this monotonicity with (51) gives

$$c_h^{\mathcal{G}}(v) = \Phi_h\left(\{c_{Q,h}^{\mathcal{G}}(v,a)\}_{a\in\mathcal{A}_s}\right) \leq \Phi_h\left(\{c_{Q,h}^{\mathcal{T}}(u,a)\}_{a\in\mathcal{A}_s}\right) = c_h^{\mathcal{T}}(u).$$

This proves the induction step. Since the base case at depth H holds, backward induction gives

$$c_h^{\mathcal{G}}(v) \leq c_h^{\mathcal{T}}(u), \qquad \forall h,\ \forall v \in \mathcal{N}_h,\ \forall u \in \phi_h^{-1}(v). \tag{52}$$

Taking $h = 0$ gives the root comparison

$$c_0^{\mathcal{G}} \leq c_0^{\mathcal{T}}.$$

Full proof-bound comparison. Combining the constant comparison with the visit-count comparison gives the corresponding comparison for the full polynomial concentration bound. Namely, for every $u \in \mathcal{C}(v)$,

$$c_h^{\mathcal{G}}(v)\big(T_v^{\mathcal{G}}(n)\big)^{-\alpha_h}\varepsilon^{-\beta_h} \le c_h^{\mathcal{T}}(u)\big(T_v^{\mathcal{G}}(n)\big)^{-\alpha_h}\varepsilon^{-\beta_h} \tag{53}$$

$$\le c_h^{\mathcal{T}}(u)\big(T_u^{\mathcal{T}}(n)\big)^{-\alpha_h}\varepsilon^{-\beta_h}, \tag{54}$$

where the first inequality uses (52), and the second uses $T_v^{\mathcal{G}}(n) \ge T_u^{\mathcal{T}}(n)$. Therefore the graph representation cannot produce a larger same-trajectory proof bound than any corresponding tree copy.

Finally, (48) gives the stated average sample-sharing factor $1/\tau_h$. This completes the proof. □

D.4 Why Depth Augmentation Is Necessary: The Cross-Depth Merging Bias

A natural question is whether we can go further and merge nodes *across different depths*: treat $(s, 3)$ and $(s, 5)$ as the same node since they correspond to the same physical state. We show that this introduces an irreducible bias that fundamentally prevents convergence to the correct value.

Proposition 3 (Cross-depth merging can introduce irreducible bias)**.** *Consider a physical state s that is visited at two depths $h_1 < h_2$ under a fixed finite horizon $H < \infty$. Let*

$$\widetilde{V}(s, h) = (\mathcal{B}^{H-h}V_0)(s)$$

be the depth-h truncated Bellman target. A full-merge estimator that stores a single statistic $\widehat{V}(s)$ for both depths generally does not target either $\widetilde{V}(s, h_1)$ or $\widetilde{V}(s, h_2)$ individually.

More precisely, suppose the empirical fractions of updates arriving from depths h_1 and h_2 converge to positive limits

$$\frac{n_{h_i}}{n_{h_1} + n_{h_2}} \longrightarrow \rho_i, \qquad \rho_i > 0, \qquad \rho_1 + \rho_2 = 1.$$

Then the limiting target of the shared statistic is the convex mixture

$$\bar{V}(s) = \rho_1\widetilde{V}(s, h_1) + \rho_2\widetilde{V}(s, h_2). \tag{55}$$

Consequently,

$$|\bar{V}(s) - \widetilde{V}(s, h_1)| = \rho_2|\widetilde{V}(s, h_2) - \widetilde{V}(s, h_1)|, \tag{56}$$

and similarly

$$|\bar{V}(s) - \widetilde{V}(s, h_2)| = \rho_1|\widetilde{V}(s, h_2) - \widetilde{V}(s, h_1)|. \tag{57}$$

Thus, whenever $\widetilde{V}(s, h_1) \neq \widetilde{V}(s, h_2)$ and both depths have nonzero asymptotic mass, the bias does not vanish as the number of visits grows.

Moreover, the possible size of this local bias is bounded by the cross-depth gap:

$$|\bar{V}(s) - \widetilde{V}(s, h_i)| \le \delta(s), \qquad i \in \{1, 2\}, \tag{58}$$

where $\delta(s)$ satisfies the bound in Lemma 3.

Proof. The mixture expression (55) follows from the law of large numbers for a shared average whose depth-h_i updates have mean $\widetilde{V}(s, h_i)$ and whose empirical depth proportions converge to ρ_i. Subtracting $\widetilde{V}(s, h_1)$ gives

$$\bar{V}(s) - \widetilde{V}(s, h_1) = \rho_1\widetilde{V}(s, h_1) + \rho_2\widetilde{V}(s, h_2) - \widetilde{V}(s, h_1) \tag{59}$$

$$= \rho_2\left(\widetilde{V}(s, h_2) - \widetilde{V}(s, h_1)\right), \tag{60}$$

which proves (56). The corresponding identity for depth h_2 is identical.

The upper bound (58) follows immediately from the definition

$$\delta(s) = \max_{h,h' \in \mathcal{D}(s)} |\widetilde{V}(s, h) - \widetilde{V}(s, h')|.$$

It remains to show that unequal depth-specific targets occur in general. Consider the one-state MDP with a single action, deterministic self-loop, reward $R(s, a, s) = 1$, discount $\gamma \in (0, 1)$, and rollout value $V_0(s) = 0$. Then

$$V^\star(s) = \frac{1}{1-\gamma}, \qquad V_0(s) \neq V^\star(s).$$

For remaining horizon $H - h$, the truncated value is

$$\widetilde{V}(s,h) = \sum_{t=0}^{H-h-1} \gamma^t = \frac{1-\gamma^{H-h}}{1-\gamma}.$$

Therefore, for $h_1 < h_2$,

$$\widetilde{V}(s,h_1) - \widetilde{V}(s,h_2) = \frac{1-\gamma^{H-h_1}}{1-\gamma} - \frac{1-\gamma^{H-h_2}}{1-\gamma} \tag{61}$$

$$= \frac{\gamma^{H-h_2} - \gamma^{H-h_1}}{1-\gamma} > 0. \tag{62}$$

Hence, if both depths are assigned positive limiting weights, the full-merge statistic has a non-vanishing bias relative to each depth-specific target. □

Remark 9 (When cross-depth merging is safe). Cross-depth merging has zero local bias in several special cases.

First, if a physical state is visited at only one depth, then $\mathcal{D}(s)$ is a singleton and $\delta(s) = 0$.

Second, if the rollout value is Bellman-consistent, $\mathcal{B}V_0 = V_0$, then Lemma 3 gives $\delta(s) = 0$ for every state. In particular, this holds when $V_0 = V^\star$.

Third, if the empirical depth weights become degenerate, meaning all asymptotic mass at state s comes from a single depth, then the shared estimator targets that depth's value even if other depths are possible but visited only negligibly.

Outside these cases, full-state merging may average samples with different depth-specific targets. The depth-augmented representation avoids this issue by keying nodes as (s,h), so every stored value estimate has a single well-defined truncated Bellman target.

Remark 10 (Practical variant: cross-depth merging with depth-dependent correction). In practice, one could merge across depths and apply a correction. If node s is visited at depths $h_1, \ldots, h_m$, maintain depth-specific running averages and use a depth-specific estimate when backpropagating from depth h. This is equivalent to maintaining per-depth statistics within a single graph node, and is an implementation optimization that is analytically equivalent to our depth-augmented formulation. We prefer the depth-augmented formulation for clarity of exposition and analysis.

Lemma 3 (Cross-depth gap bound)**.** *Assume $\gamma \in (0,1)$ and let*

$$\widetilde{V}(s,h) = (\mathcal{B}^{H-h}V_0)(s)$$

be the depth-h truncated Bellman value. For any state s and any two depths $h_1 < h_2$, we have

$$\left|\widetilde{V}(s,h_1) - \widetilde{V}(s,h_2)\right| \le \frac{\gamma^{H-h_2} - \gamma^{H-h_1}}{1-\gamma} \|\mathcal{B}V_0 - V_0\|_\infty. \tag{63}$$

Consequently, for

$$\delta(s) = \max_{h_1,h_2 \in \mathcal{D}(s)} |\widetilde{V}(s,h_1) - \widetilde{V}(s,h_2)|,$$

we have

$$\delta(s) \le \frac{\gamma^{H-h_{\max}(s)} - \gamma^{H-h_{\min}(s)}}{1-\gamma} \|\mathcal{B}V_0 - V_0\|_\infty. \tag{64}$$

Moreover,

$$\delta(s) \le \frac{(1+\gamma)\left(\gamma^{H-h_{\max}(s)} - \gamma^{H-h_{\min}(s)}\right)}{1-\gamma} \|V^\star - V_0\|_\infty. \tag{65}$$

Proof. The case $h_1 = h_2$ is trivial, so assume $h_1 < h_2$. Let

$$k_1 = H - h_1, \qquad k_2 = H - h_2, \qquad d = k_1 - k_2 = h_2 - h_1.$$

Then $k_1 > k_2$ and

$$\widetilde{V}(\cdot,h_1) = \mathcal{B}^{k_1}V_0 = \mathcal{B}^{k_2}(\mathcal{B}^d V_0), \qquad \widetilde{V}(\cdot,h_2) = \mathcal{B}^{k_2}V_0.$$

Since $\mathcal{B}$ is a γ-contraction in sup norm,

$$\|\widetilde{V}(\cdot,h_1) - \widetilde{V}(\cdot,h_2)\|_\infty = \left\|\mathcal{B}^{k_2}(\mathcal{B}^d V_0) - \mathcal{B}^{k_2}V_0\right\|_\infty \tag{66}$$

$$\le \gamma^{k_2} \left\|\mathcal{B}^d V_0 - V_0\right\|_\infty. \tag{67}$$

Now use the telescoping decomposition

$$\mathcal{B}^d V_0 - V_0 = \sum_{j=0}^{d-1} \left(\mathcal{B}^{j+1} V_0 - \mathcal{B}^j V_0\right).$$

Again by contraction,

$$\left\|\mathcal{B}^{j+1} V_0 - \mathcal{B}^j V_0\right\|_\infty \le \gamma^j \|\mathcal{B} V_0 - V_0\|_\infty.$$

Therefore,

$$\left\|\mathcal{B}^d V_0 - V_0\right\|_\infty \le \sum_{j=0}^{d-1} \gamma^j \|\mathcal{B} V_0 - V_0\|_\infty \tag{68}$$

$$= \frac{1-\gamma^d}{1-\gamma} \|\mathcal{B} V_0 - V_0\|_\infty. \tag{69}$$

Combining the last two displays gives

$$\|\widetilde{V}(\cdot, h_1) - \widetilde{V}(\cdot, h_2)\|_\infty \le \frac{\gamma^{k_2} - \gamma^{k_2+d}}{1-\gamma} \|\mathcal{B} V_0 - V_0\|_\infty.$$

Since $k_2 = H - h_2$ and $k_2 + d = H - h_1$, this is exactly

$$\|\widetilde{V}(\cdot, h_1) - \widetilde{V}(\cdot, h_2)\|_\infty \le \frac{\gamma^{H-h_2} - \gamma^{H-h_1}}{1-\gamma} \|\mathcal{B} V_0 - V_0\|_\infty.$$

Taking the maximum over $h_1, h_2 \in \mathcal{D}(s)$ gives (64), because the right-hand side is maximized by $h_1 = h_{\min}(s)$ and $h_2 = h_{\max}(s)$.

Finally, since $V^\star = \mathcal{B} V^\star$,

$$\|\mathcal{B} V_0 - V_0\|_\infty = \|(\mathcal{B} V_0 - \mathcal{B} V^\star) + (V^\star - V_0)\|_\infty \tag{70}$$

$$\le \|\mathcal{B} V_0 - \mathcal{B} V^\star\|_\infty + \|V^\star - V_0\|_\infty \tag{71}$$

$$\le (1+\gamma) \|V^\star - V_0\|_\infty. \tag{72}$$

Substituting this into (64) proves (65). □

Definition 4 (Empirical effective target for GS-Power-UCT-F)**.** *Let $T_{s,a,h}(n)$ denote the number of times action a was selected at physical state s when entered at trajectory depth h during the first n simulations, and write $T_{s,a}(n) = \sum_h T_{s,a,h}(n)$. Define the depth-mixture weights $\omega_{n,h}(s,a) = T_{s,a,h}(n)/T_{s,a}(n)$ when $T_{s,a}(n) > 0$, and arbitrarily on the simplex otherwise. For $V : \mathcal{S} \to \mathbb{R}$, let $B_h V(s') = V_0(s')$ if $h + 1 = H$ and $V(s')$ otherwise. The empirical full-merge Bellman operator is*

$$(\mathcal{T}_{F,n}^H V)(s) = \max_{a \in \mathcal{A}_s} \sum_{h=0}^{H-1} \omega_{n,h}(s,a) \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma B_h V(s')\right], \tag{73}$$

which is a γ-contraction; we write $\bar{V}_n$ for its unique fixed point and define the corresponding Q-target

$$\bar{Q}_n(s,a) = \sum_{h=0}^{H-1} \omega_{n,h}(s,a) \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma B_h \bar{V}_n(s')\right]. \tag{74}$$

D.5 Proof of Theorem 3

Before proving Theorem 3, we establish the following results.

Lemma 4 (Robust Q-propagation under full merging)**.** *Fix a physical state-action pair (s,a) in the full-merge algorithm and consider the first $t \ge 1$ updates to the shared statistic $\widehat{Q}(s,a)$, where the i-th update arrives when s is entered at depth h_i with sampled successor $S_i \sim P(\cdot \mid s,a)$, reward R_i, and recursive child return W_i from $(S_i, h_i + 1)$. For any reference depth h, let $\widetilde{Q}(s,h,a) = \sum_{s'} P(s' \mid s,a)[R(s,a,s') + \gamma \widetilde{V}(s', h+1)]$ and define the cross-depth drift $\delta_{Q,t}(s,a;h) := \max_{1 \le i \le t} |\widetilde{Q}(s,h_i,a) - \widetilde{Q}(s,h,a)|$. Assume rewards and value estimates are*

bounded with reward range $\leq L_R$ *and* $|W_i|, |\widetilde{V}(S_i, h_i + 1)| \leq L_V$ *a.s., and that there exist constants* $B \geq 0$, $c_V > 0$, $\alpha > 0$, $\beta > 1$ *with* $\rho := \alpha/\beta \in (0, 1)$ *such that for every* i *and* $u > 0$,

$$\mathbb{P}\left(\left|W_i - \widetilde{V}(S_i, h_i + 1)\right| > B + u \middle| \mathcal{F}_{i,1}\right) \leq c_V J_i^{-\alpha} u^{-\beta}, \tag{75}$$

where $\mathcal{F}_{i,1}$ *is the history after observing* (S_i, R_i) *but before the recursive child randomness, and* $J_i := 1 + \#\{r < i : (S_r, h_r) = (S_i, h_i)\}$ *is the local occurrence count of the same successor-depth pair among the first* i *updates. Then there exist constants* $\kappa, \sigma_Q < \infty$, *independent of* t, *such that for every* $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\widehat{Q}_t(s, a) - \widetilde{Q}(s, h, a)\right| > \delta_{Q,t}(s, a; h) + \gamma B + \gamma\kappa t^{-\rho} + \varepsilon\right) \leq 2\exp\left(-\tfrac{t\varepsilon^2}{2\sigma_Q^2}\right). \tag{76}$$

Consequently, for any $\bar{\alpha}, \bar{\beta}$ *with* $\bar{\beta} > 1$, $2\bar{\alpha} \leq \bar{\beta}$, *and* $\bar{\alpha} \leq \rho\bar{\beta}$, *there exists* $C_Q < \infty$ *independent of* t *such that*

$$\mathbb{P}\left(\left|\widehat{Q}_t(s, a) - \widetilde{Q}(s, h, a)\right| > \delta_{Q,t}(s, a; h) + \gamma B + \varepsilon\right) \leq C_Q t^{-\bar{\alpha}} \varepsilon^{-\bar{\beta}}; \tag{77}$$

in particular, if $B = B_\Delta \geq \delta_{Q,t}(s, a; h)/(1 - \gamma)$, *then*

$$\mathbb{P}\left(\left|\widehat{Q}_t(s, a) - \widetilde{Q}(s, h, a)\right| > B_\Delta + \varepsilon\right) \leq C_Q t^{-\bar{\alpha}} \varepsilon^{-\bar{\beta}}. \tag{78}$$

Proof. Let

$$q_i := \widetilde{Q}(s, h_i, a), \qquad q_h := \widetilde{Q}(s, h, a), \qquad v_i := \widetilde{V}(S_i, h_i + 1).$$

The i-th backup satisfies

$$Y_i - q_h = \underbrace{[R_i + \gamma v_i - q_i]}_{A_i} + \gamma \underbrace{[W_i - v_i]}_{e_i} + \underbrace{[q_i - q_h]}_{d_i}. \tag{79}$$

Averaging over $i = 1, \ldots, t$, we obtain

$$\widehat{Q}_t(s, a) - q_h = \frac{1}{t}\sum_{i=1}^{t} A_i + \frac{\gamma}{t}\sum_{i=1}^{t} e_i + \frac{1}{t}\sum_{i=1}^{t} d_i. \tag{80}$$

We bound the three terms separately.

First, by definition of $\delta_{Q,t}(s, a; h)$,

$$|d_i| = \left|\widetilde{Q}(s, h_i, a) - \widetilde{Q}(s, h, a)\right| \leq \delta_{Q,t}(s, a; h),$$

and therefore

$$\left|\frac{1}{t}\sum_{i=1}^{t} d_i\right| \leq \delta_{Q,t}(s, a; h). \tag{81}$$

Second, consider the transition-reward term

$$A_i = R_i + \gamma\widetilde{V}(S_i, h_i + 1) - \widetilde{Q}(s, h_i, a).$$

Let $\mathcal{F}_{i,0}$ be the history immediately before drawing (S_i, R_i) on the i-th update to (s, a). Conditional on $\mathcal{F}_{i,0}$, the depth h_i is fixed, and

$$\mathbb{E}\left[R_i + \gamma\widetilde{V}(S_i, h_i + 1) \middle| \mathcal{F}_{i,0}\right] = \widetilde{Q}(s, h_i, a).$$

Hence

$$\mathbb{E}[A_i \mid \mathcal{F}_{i,0}] = 0. \tag{82}$$

Thus A_i is a martingale-difference term. Since the reward range is at most L_R and $|\widetilde{V}| \leq L_V$, there exists a finite constant

$$L_A := L_R + 2\gamma L_V$$

such that

$$|A_i| \leq L_A$$

almost surely.

Third, decompose the child-estimation error

$$e_i = W_i - v_i$$

into its predictable and centered parts. Define

$$b_i := \mathbb{E}[e_i \mid \mathcal{F}_{i,1}], \qquad \xi_i := e_i - b_i.$$

Then

$$\mathbb{E}[\xi_i \mid \mathcal{F}_{i,1}] = 0. \tag{83}$$

Moreover, because $|W_i| \le L_V$ and $|v_i| \le L_V$,

$$|e_i| \le 2L_V.$$

Thus

$$|\xi_i| = |e_i - \mathbb{E}[e_i \mid \mathcal{F}_{i,1}]| \le 4L_V.$$

Set

$$L_\xi := 4L_V.$$

We now bound the predictable child bias b_i. By Jensen's inequality,

$$|b_i| \le \mathbb{E}[|e_i| \mid \mathcal{F}_{i,1}].$$

Also,

$$|e_i| \le B + \big(|e_i| - B\big)_+.$$

Therefore,

$$|b_i| \le B + \mathbb{E}\left[\big(|e_i| - B\big)_+ \middle| \mathcal{F}_{i,1}\right]. \tag{84}$$

Using the tail-integral formula,

$$\mathbb{E}\left[\big(|e_i| - B\big)_+ \middle| \mathcal{F}_{i,1}\right] = \int_0^\infty \mathbb{P}\left(|e_i| > B + u | \mathcal{F}_{i,1}\right) du. \tag{85}$$

Let

$$\eta_i := J_i^{-\alpha/\beta} = J_i^{-\rho}.$$

Splitting the integral at η_i and using condition (75), we get

$$\begin{aligned}\int_0^\infty \mathbb{P}\left(|e_i| > B + u | \mathcal{F}_{i,1}\right) du &\le \eta_i + \int_{\eta_i}^\infty c_V J_i^{-\alpha} u^{-\beta}\, du \\ &= \eta_i + \frac{c_V}{\beta - 1} J_i^{-\alpha} \eta_i^{1-\beta}.\end{aligned}$$

Since $\eta_i = J_i^{-\alpha/\beta}$,

$$J_i^{-\alpha} \eta_i^{1-\beta} = J_i^{-\alpha} J_i^{-(\alpha/\beta)(1-\beta)} = J_i^{-\alpha/\beta} = J_i^{-\rho}.$$

Hence

$$\mathbb{E}\left[\big(|e_i| - B\big)_+ \middle| \mathcal{F}_{i,1}\right] \le \left(1 + \frac{c_V}{\beta - 1}\right) J_i^{-\rho}.$$

Define

$$C_{\text{ch}} := 1 + \frac{c_V}{\beta - 1}.$$

Then

$$|b_i| \le B + C_{\text{ch}} J_i^{-\rho}. \tag{86}$$

It remains to average the local occurrence factors $J_i^{-\rho}$. The possible groups are successor-depth pairs

$$g = (s', \ell),$$

where s' is a possible successor of (s, a), and $\ell \in \{0, \dots, H-1\}$ is a possible parent depth. Let G be the number of such groups. Since the successor support of (s, a) is finite and the horizon is finite,

$$G < \infty.$$

Let $n_g(t)$ be the number of indices $i \le t$ belonging to group g. For a fixed group g, the corresponding values of J_i are exactly

$$1, 2, \dots, n_g(t).$$

Therefore,

$$\sum_{i=1}^{t} J_i^{-\rho} = \sum_{g} \sum_{j=1}^{n_g(t)} j^{-\rho}. \tag{87}$$

Since $\rho \in (0,1)$,

$$\sum_{j=1}^{n} j^{-\rho} \leq 1 + \int_1^n x^{-\rho}\, dx \leq \left(1 + \frac{1}{1-\rho}\right) n^{1-\rho}.$$

Let

$$C_\rho := 1 + \frac{1}{1-\rho}.$$

Then

$$\sum_{i=1}^{t} J_i^{-\rho} \leq C_\rho \sum_{g} n_g(t)^{1-\rho}. \tag{88}$$

By concavity of $x \mapsto x^{1-\rho}$,

$$\sum_{g} n_g(t)^{1-\rho} \leq G^\rho \left(\sum_{g} n_g(t)\right)^{1-\rho}.$$

Since $\sum_g n_g(t) = t$,

$$\sum_{i=1}^{t} J_i^{-\rho} \leq C_\rho G^\rho t^{1-\rho}. \tag{89}$$

Dividing by t,

$$\frac{1}{t}\sum_{i=1}^{t} J_i^{-\rho} \leq C_\rho G^\rho t^{-\rho}. \tag{90}$$

Combining (11) and (15), we obtain

$$\left|\frac{1}{t}\sum_{i=1}^{t} b_i\right| \leq \frac{1}{t}\sum_{i=1}^{t} |b_i| \leq B + C_{\mathrm{ch}} C_\rho G^\rho t^{-\rho}.$$

Define

$$\kappa := C_{\mathrm{ch}} C_\rho G^\rho.$$

Then

$$\left|\frac{\gamma}{t}\sum_{i=1}^{t} b_i\right| \leq \gamma B + \gamma\kappa t^{-\rho}. \tag{91}$$

Using $e_i = b_i + \xi_i$, equation (5) becomes

$$\widehat{Q}_t(s,a) - q_h = \frac{1}{t}\sum_{i=1}^{t} [A_i + \gamma\xi_i] + \frac{\gamma}{t}\sum_{i=1}^{t} b_i + \frac{1}{t}\sum_{i=1}^{t} d_i. \tag{92}$$

The last two terms are controlled by (6) and (16). It remains to control the martingale term

$$M_t := \sum_{i=1}^{t} [A_i + \gamma\xi_i].$$

Consider the refined filtration

$$\mathcal{F}_{i,0} \subset \mathcal{F}_{i,1} \subset \mathcal{F}_{i,2},$$

where $\mathcal{F}_{i,0}$ is the history before drawing (S_i, R_i), $\mathcal{F}_{i,1}$ is the history after drawing (S_i, R_i), and $\mathcal{F}_{i,2}$ is the history after completing the recursive child call. The term A_i is a martingale difference from $\mathcal{F}_{i,0}$ to $\mathcal{F}_{i,1}$, and $\gamma\xi_i$ is a martingale difference from $\mathcal{F}_{i,1}$ to $\mathcal{F}_{i,2}$. Their absolute bounds are L_A and γL_ξ, respectively.

Therefore, by Azuma-Hoeffding applied to this refined martingale,

$$\mathbb{P}\left(|M_t| > t\varepsilon\right) \leq 2\exp\left(-\frac{t^2\varepsilon^2}{2t(L_A^2 + \gamma^2 L_\xi^2)}\right).$$

Define

$$\sigma_Q^2 := L_A^2 + \gamma^2 L_\xi^2.$$

Then

$$\mathbb{P}\left(\left|\frac{1}{t}M_t\right| > \varepsilon\right) \le 2\exp\left(-\frac{t\varepsilon^2}{2\sigma_Q^2}\right). \tag{93}$$

Combining (81), (91), (92), and (93), we obtain

$$\mathbb{P}\left(\left|\widehat{Q}_t(s,a) - q_h\right| > \delta_{Q,t}(s,a;h) + \gamma B + \gamma\kappa t^{-\rho} + \varepsilon\right) \le 2\exp\left(-\frac{t\varepsilon^2}{2\sigma_Q^2}\right),$$

which proves (76).

We now convert (76) into the polynomial bound (77). Fix $\bar{\alpha}, \bar{\beta}$ satisfying

$$\bar{\beta} > 1, \qquad 2\bar{\alpha} \le \bar{\beta}, \qquad \bar{\alpha} \le \rho\bar{\beta}.$$

There are two cases.

First suppose

$$0 < \varepsilon \le 2\gamma\kappa t^{-\rho}.$$

Then the probability in (77) is at most 1. Moreover,

$$t^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}} \ge t^{-\bar{\alpha}}\left(2\gamma\kappa t^{-\rho}\right)^{-\bar{\beta}} = (2\gamma\kappa)^{-\bar{\beta}} t^{-\bar{\alpha}+\rho\bar{\beta}}.$$

Since $\bar{\alpha} \le \rho\bar{\beta}$, we have $t^{-\bar{\alpha}+\rho\bar{\beta}} \ge 1$. Hence, choosing

$$C_Q \ge (2\gamma\kappa)^{\bar{\beta}}$$

makes

$$1 \le C_Q t^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}.$$

Now suppose

$$\varepsilon > 2\gamma\kappa t^{-\rho}.$$

Then

$$\gamma\kappa t^{-\rho} < \frac{\varepsilon}{2}.$$

Therefore,

$$\begin{aligned}&\left\{\left|\widehat{Q}_t(s,a) - q_h\right| > \delta_{Q,t}(s,a;h) + \gamma B + \varepsilon\right\} \\ &\quad\subseteq \left\{\left|\widehat{Q}_t(s,a) - q_h\right| > \delta_{Q,t}(s,a;h) + \gamma B + \gamma\kappa t^{-\rho} + \frac{\varepsilon}{2}\right\}.\end{aligned}$$

Applying (1) with $\varepsilon/2$ gives

$$\mathbb{P}\left(\left|\widehat{Q}_t(s,a) - q_h\right| > \delta_{Q,t}(s,a;h) + \gamma B + \varepsilon\right) \le 2\exp\left(-\frac{t\varepsilon^2}{8\sigma_Q^2}\right). \tag{94}$$

For every $c > 0$ and every $\bar{\beta} > 0$, there exists $K(c,\bar{\beta}) < \infty$ such that

$$e^{-cx} \le K(c,\bar{\beta})x^{-\bar{\beta}/2} \qquad \forall x > 0.$$

Using $x = t\varepsilon^2$ and $c = 1/(8\sigma_Q^2)$, (19) implies

$$2\exp\left(-\frac{t\varepsilon^2}{8\sigma_Q^2}\right) \le 2Kt^{-\bar{\beta}/2}\varepsilon^{-\bar{\beta}}.$$

Since $2\bar{\alpha} \le \bar{\beta}$,

$$t^{-\bar{\beta}/2} \le t^{-\bar{\alpha}}.$$

Thus

$$2\exp\left(-\frac{t\varepsilon^2}{8\sigma_Q^2}\right) \le 2Kt^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}.$$

Increasing C_Q if necessary proves (2).

Finally, suppose $B = B_\Delta$ and

$$B_\Delta \geq \frac{\delta_{Q,t}(s,a;h)}{1-\gamma}.$$

Then

$$\delta_{Q,t}(s,a;h) \leq (1-\gamma)B_\Delta.$$

Therefore,

$$\delta_{Q,t}(s,a;h) + \gamma B_\Delta \leq (1-\gamma)B_\Delta + \gamma B_\Delta = B_\Delta.$$

Hence

$$\begin{aligned}&\left\{\left|\widehat{Q}_t(s,a) - \widetilde{Q}(s,h,a)\right| > B_\Delta + \varepsilon\right\}\\ &\quad\subseteq \left\{\left|\widehat{Q}_t(s,a) - \widetilde{Q}(s,h,a)\right| > \delta_{Q,t}(s,a;h) + \gamma B_\Delta + \varepsilon\right\}.\end{aligned}$$

Applying (77) yields

$$\mathbb{P}\left(\left|\widehat{Q}_t(s,a) - \widetilde{Q}(s,h,a)\right| > B_\Delta + \varepsilon\right) \leq C_Q t^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}.$$

This proves (78), completing the proof. □

Corollary 1 (Control by the cross-depth gap)**.** *Suppose the state-value cross-depth gap satisfies*

$$\Delta_{\text{CROSS}} \geq \max_{s'}\max_{h,k}\left|\widetilde{V}(s',h+1) - \widetilde{V}(s',k+1)\right|$$

over the successor states and depths relevant to the updates of (s,a). Then

$$\delta_{Q,t}(s,a;h) \leq \gamma\Delta_{\text{CROSS}}.$$

Consequently, one may choose

$$B_\Delta = \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}},$$

and Lemma 4 gives

$$\mathbb{P}\left(\left|\widehat{Q}_t(s,a) - \widetilde{Q}(s,h,a)\right| > \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}} + \varepsilon\right) \leq C_Q t^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}.$$

Proof. For any two depths h and k,

$$\begin{aligned}\left|\widetilde{Q}(s,h,a) - \widetilde{Q}(s,k,a)\right| &= \gamma\left|\sum_{s'} P(s' \mid s,a)\left[\widetilde{V}(s',h+1) - \widetilde{V}(s',k+1)\right]\right|\\ &\leq \gamma\max_{s'}\left|\widetilde{V}(s',h+1) - \widetilde{V}(s',k+1)\right|\\ &\leq \gamma\Delta_{\text{CROSS}}.\end{aligned}$$

Taking the maximum over the depths appearing in the first t updates gives $\delta_{Q,t}(s,a;h) \leq \gamma\Delta_{\text{CROSS}}$. The claimed choice of B_Δ then satisfies

$$B_\Delta = \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}} \geq \frac{\delta_{Q,t}(s,a;h)}{1-\gamma},$$

so the result follows from Lemma 4. □

Lemma 5 (Robust Power-UCT value step)**.** *Fix a physical state s, reference depth h, and finite action set $\mathcal{A}_s$, writing $q_a := \widetilde{Q}(s,h,a)$ and $v_\star := \widetilde{V}(s,h) = \max_{a\in\mathcal{A}_s} q_a$. After t visits to the full-merge node s, let $T_{s,a}(t)$ count selections of a and $\widehat{Q}_m(s,a)$ be the shared Q-estimate after m updates to (s,a), so the value estimate is the power mean*

$$\widehat{V}_t(s) = \left(\sum_{a\in\mathcal{A}_s}\frac{T_{s,a}(t)}{t}\big(\widehat{Q}_{T_{s,a}(t)}(s,a)\big)^p\right)^{1/p}, \qquad p \in [1,\infty).$$

Assume all Q-estimates and targets are nonnegative and bounded by L, and (after the usual forced initialization) the local action-selection rule uses a single Power-UCT triple (b,α_+,β_+):

$$a_j \in \arg\max_{a\in\mathcal{A}_s}\left\{\widehat{Q}_{T_{s,a}(j-1)}(s,a) + C\,\frac{T_s(j-1)^{b/\beta_+}}{T_{s,a}(j-1)^{\alpha_+/\beta_+}}\right\}.$$

*Suppose there exist effective targets $\{\bar{q}_a\}_{a\in\mathcal{A}_s}$ and a bias radius $B \geq 0$ with $\max_{a\in\mathcal{A}_s} |\bar{q}_a - q_a| \leq B$ **(H-V)**, and that for every a and every $m \geq 1, \varepsilon > 0$,*

$$\mathbb{P}(|\widehat{Q}_m(s,a) - \bar{q}_a| > \varepsilon) \leq c_Q m^{-\alpha_+} \varepsilon^{-\beta_+}. \tag{95}$$

Assume further the Power-UCT parameter conditions $2 < b < \alpha_+$ and either ($1 \leq p \leq 2$, $\alpha_+ \leq \beta_+/2$) or ($p > 2$, $\alpha_+ \leq \beta_+/2$, $0 < \alpha_+ - \beta_+/p < 1$), and let $\alpha_V := (b-1)(1-b/\alpha_+)$ and $\beta_V := b - 1$. Then there exists $C_V < \infty$, depending only on $K = |\mathcal{A}_s|, p, C, L, c_Q, b, \alpha_+, \beta_+$ but not on t, ε, B, such that

$$\mathbb{P}(|\widehat{V}_t(s) - \widetilde{V}(s,h)| > B + \varepsilon) \leq C_V t^{-\alpha_V} \varepsilon^{-\beta_V}, \qquad \forall t \geq 1,\ \varepsilon > 0. \tag{96}$$

Proof. Define the effective value target induced by the effective action targets $(\bar{q}_a)_{a\in\mathcal{A}_s}$ as

$$\bar{v} := \max_{a\in\mathcal{A}_s} \bar{q}_a.$$

The standard Power-UCT concentration theorem says the following. If, at a node, each action-level estimator $\widehat{Q}_m(s,a)$ satisfies

$$\mathbb{P}\left(|\widehat{Q}_m(s,a) - \bar{q}_a| > \varepsilon\right) \leq c_Q m^{-\alpha_+} \varepsilon^{-\beta_+},$$

and the node uses the Power-UCT selection rule with parameters satisfying the conditions above, then the power-mean value estimator satisfies

$$\mathbb{P}\left(|\widehat{V}_t(s) - \bar{v}| > \varepsilon\right) \leq C_V t^{-\alpha_V} \varepsilon^{-\beta_V}, \qquad \forall t \geq 1,\ \forall \varepsilon > 0, \tag{97}$$

where

$$\alpha_V = (b-1)\left(1 - \frac{b}{\alpha_+}\right), \qquad \beta_V = b - 1.$$

This theorem is purely local: it only requires action-level Q-concentration and the Power-UCT action-selection rule at the node. It does not depend on whether the Q-estimates were produced by a tree, a depth-augmented graph, or a full-merge graph.

It remains to transfer concentration from the effective target $\bar{v}$ to the depth-specific truncated target $v_\star = \widetilde{V}(s,h)$. By definition,

$$v_\star = \max_{a\in\mathcal{A}_s} q_a, \qquad \bar{v} = \max_{a\in\mathcal{A}_s} \bar{q}_a.$$

The max map is 1-Lipschitz in the sup norm. Therefore,

$$\begin{aligned} |\bar{v} - v_\star| &= \left|\max_{a\in\mathcal{A}_s} \bar{q}_a - \max_{a\in\mathcal{A}_s} q_a\right| \\ &\leq \max_{a\in\mathcal{A}_s} |\bar{q}_a - q_a| \\ &\leq B, \end{aligned} \tag{98}$$

where the last inequality is hypothesis (H-V).

Now suppose

$$|\widehat{V}_t(s) - v_\star| > B + \varepsilon.$$

Using the triangle inequality and (98),

$$|\widehat{V}_t(s) - \bar{v}| \geq |\widehat{V}_t(s) - v_\star| - |\bar{v} - v_\star| > B + \varepsilon - B = \varepsilon.$$

Hence the event inclusion

$$\left\{|\widehat{V}_t(s) - v_\star| > B + \varepsilon\right\} \subseteq \left\{|\widehat{V}_t(s) - \bar{v}| > \varepsilon\right\}$$

holds. Applying the Stochastic-Power-UCT bound (97) gives

$$\mathbb{P}\left(|\widehat{V}_t(s) - v_\star| > B + \varepsilon\right) \leq \mathbb{P}\left(|\widehat{V}_t(s) - \bar{v}| > \varepsilon\right) \leq C_V t^{-\alpha_V} \varepsilon^{-\beta_V}.$$

Since $v_\star = \widetilde{V}(s,h)$, this is exactly (96). □

Corollary 2 (Value-step bias controlled by Q-level cross-depth drift)**.** *Under the assumptions of Lemma 5, suppose*

$$\max_{a\in\mathcal{A}_s} |\bar{q}_a - \widetilde{Q}(s,h,a)| \le B_\Delta.$$

Then

$$\mathbb{P}\left(|\widehat{V}_t(s) - \widetilde{V}(s,h)| > B_\Delta + \varepsilon\right) \le C_V t^{-\alpha_V}\varepsilon^{-\beta_V}.$$

In particular, if the Q-level effective targets satisfy

$$\max_{a\in\mathcal{A}_s} |\bar{q}_a - \widetilde{Q}(s,h,a)| \le \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}},$$

then

$$\mathbb{P}\left(|\widehat{V}_t(s) - \widetilde{V}(s,h)| > \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}} + \varepsilon\right) \le C_V t^{-\alpha_V}\varepsilon^{-\beta_V}.$$

Proof. Apply Lemma 5 with

$$B = B_\Delta.$$

The second claim follows from the specific choice

$$B_\Delta = \frac{\gamma}{1-\gamma}\Delta_{\text{CROSS}}.$$

□

Proof of Theorem 3

Proof. We prove a stronger biased concentration statement and then integrate its tail.

For a physical state s and depth h, recall that

$$\widetilde{V}(s,h) = (\mathcal{B}^{H-h}V_0)(s),$$

and

$$\widetilde{Q}(s,h,a) = \sum_{s'} P(s' \mid s,a)\left[R(s,a,s') + \gamma\widetilde{V}(s',h+1)\right].$$

For the first n simulations, define the uniform Q-level cross-depth drift

$$\delta_{Q,n} \triangleq \max_{\substack{(s,a):\, T_{s,a}(n)\ge 1 \\ h\in\mathcal{D}_{H,n}(s),\ 1\le t\le T_{s,a}(n)}} \delta_{Q,t}(s,a;h),$$

with the maximum over an empty set taken as zero. Thus $\delta_{Q,n}$ uniformly bounds the drift of every state-action-depth triple realized in the search.

The key claim is that, for every depth $h \in \{0,\dots,H\}$, every physical state s, and every visit count $t \ge 1$, there exists a constant $c_h < \infty$, independent of t, n, and ε, such that

$$\mathbb{P}\left(\left|\widehat{V}_t(s) - \widetilde{V}(s,h)\right| > B_{\Delta,n} + \varepsilon\right) \le c_h t^{-\alpha_h}\varepsilon^{-\beta_h}, \qquad \forall\varepsilon > 0. \tag{99}$$

Here $\widehat{V}_t(s)$ denotes the full-merge value estimate at the physical node s after t visits to that node. Notice that the same estimator $\widehat{V}_t(s)$ may be compared with several depth-specific targets $\widetilde{V}(s,h)$, because the full-merge node s may be entered at several depths. The price of this comparison is the additive bias radius $B_{\Delta,n}$.

We prove (99) by backward induction on h.

Base case: $h = H$. At depth H, the recursive simulation stops and returns an independent rollout or evaluation sample with mean $V_0(s) = \widetilde{V}(s,H)$. Since rollout values are bounded, Hoeffding's inequality implies that, for any admissible α_H, β_H with $\beta_H > 1$ and $\alpha_H \le \beta_H/2$, there exists $c_H < \infty$ such that

$$\mathbb{P}\left(\left|\widehat{V}_t(s) - \widetilde{V}(s,H)\right| > \varepsilon\right) \le c_H t^{-\alpha_H}\varepsilon^{-\beta_H}.$$

Since $B_{\Delta,n} \ge 0$, this immediately gives

$$\mathbb{P}\left(\left|\widehat{V}_t(s) - \widetilde{V}(s,H)\right| > B_{\Delta,n} + \varepsilon\right) \le c_H t^{-\alpha_H}\varepsilon^{-\beta_H}.$$

Thus (99) holds at depth H.

Inductive step. Fix $h < H$, and assume that (99) holds for all depths $h+1, h+2, \dots, H$. We prove it at depth h.

Fix a physical state s and action a. Consider the first t updates to the shared full-merge statistic $\widehat{Q}(s,a)$. The i-th such update may arrive when s is entered at some depth h_i, not necessarily equal to the reference depth h. Let S_i be the sampled successor, R_i the sampled reward, and W_i the value returned by the recursive child call from (S_i, h_i+1). With the corrected backup,

$$Y_i = R_i + \gamma W_i, \qquad \widehat{Q}_t(s,a) = \frac{1}{t}\sum_{i=1}^{t} Y_i.$$

By the induction hypothesis applied at the child depth h_i+1, the child returned value satisfies the biased concentration condition required by Lemma 4, with bias radius $B_{\Delta,n}$. Hence, for any reference depth h, Lemma 4 yields

$$\mathbb{P}\left(\left|\widehat{Q}_t(s,a) - \widetilde{Q}(s,h,a)\right| > B_{\Delta,n} + \varepsilon\right) \le C_Q t^{-\alpha_{h+1}} \varepsilon^{-\beta_{h+1}}, \qquad \forall \varepsilon > 0. \tag{100}$$

Indeed, the only deterministic drift term in Lemma 4 is

$$\delta_{Q,t}(s,a;h) := \max_{1\le i\le t} \left|\widetilde{Q}(s,h_i,a) - \widetilde{Q}(s,h,a)\right|.$$

By definition of $\delta_{Q,n}$,

$$\delta_{Q,t}(s,a;h) \le \delta_{Q,n}.$$

Since

$$B_{\Delta,n} = \frac{\delta_{Q,n}}{1-\gamma},$$

we have

$$\delta_{Q,t}(s,a;h) + \gamma B_{\Delta,n} \le \delta_{Q,n} + \gamma\frac{\delta_{Q,n}}{1-\gamma} = \frac{\delta_{Q,n}}{1-\gamma} = B_{\Delta,n}.$$

Therefore the robust Q-propagation lemma gives exactly the biased Q-concentration bound (100).

Now apply Lemma 5 at the physical node s. The local action targets are

$$q_a = \widetilde{Q}(s,h,a), \qquad \widetilde{V}(s,h) = \max_{a\in\mathcal{A}_s} q_a.$$

By (100), every shared action estimate $\widehat{Q}(s,a)$ concentrates around its depth-h target $\widetilde{Q}(s,h,a)$ with the same additive bias radius $B_{\Delta,n}$. Since the full-merge node uses a single depth-independent Power-UCT rule, the robust Power-UCT value-step lemma applies and gives

$$\mathbb{P}\left(\left|\widehat{V}_t(s) - \widetilde{V}(s,h)\right| > B_{\Delta,n} + \varepsilon\right) \le c_h t^{-\alpha_h}\varepsilon^{-\beta_h}. \tag{101}$$

This is precisely (99) at depth h. The induction is complete.

Root concentration. Every simulation starts from s_0 at trajectory depth 0. Therefore the full-merge root visit count after n simulations satisfies

$$T_{s_0}(n) \ge n.$$

Applying (99) with $s = s_0$, $h = 0$, and $t = T_{s_0}(n)$, we obtain

$$\mathbb{P}\left(\left|\widehat{V}_n(s_0) - \widetilde{V}(s_0,0)\right| > B_{\Delta,n} + \varepsilon\right) \le c_0 n^{-\alpha_0}\varepsilon^{-\beta_0}, \qquad \forall \varepsilon > 0. \tag{102}$$

Here $\widehat{V}_n(s_0)$ denotes the root estimate after n simulations.

Tail integration. Let

$$X_n := \left|\widehat{V}_n(s_0) - \widetilde{V}(s_0,0)\right|.$$

Since $B_{\Delta,n} \ge 0$,

$$X_n \le B_{\Delta,n} + (X_n - B_{\Delta,n})_+.$$

Taking expectations and using the tail-integral identity,

$$\mathbb{E}[X_n] \le \mathbb{E}[B_{\Delta,n}] + \int_0^\infty \mathbb{P}\left(X_n > B_{\Delta,n} + \varepsilon\right) d\varepsilon.$$

Using (102),

$$\mathbb{E}[X_n] \le \mathbb{E}[B_{\Delta,n}] + \int_0^\infty \min\left\{1,\, c_0 n^{-\alpha_0} \varepsilon^{-\beta_0}\right\} d\varepsilon.$$

Set

$$\eta_n := n^{-\alpha_0/\beta_0}.$$

Then

$$\begin{aligned}\mathbb{E}[X_n] &\le \mathbb{E}[B_{\Delta,n}] + \eta_n + \int_{\eta_n}^\infty c_0 n^{-\alpha_0} \varepsilon^{-\beta_0}\, d\varepsilon \\ &= \mathbb{E}[B_{\Delta,n}] + \eta_n + \frac{c_0}{\beta_0 - 1} n^{-\alpha_0} \eta_n^{1-\beta_0}.\end{aligned}$$

Because $\eta_n = n^{-\alpha_0/\beta_0}$,

$$n^{-\alpha_0} \eta_n^{1-\beta_0} = n^{-\alpha_0} n^{-(\alpha_0/\beta_0)(1-\beta_0)} = n^{-\alpha_0/\beta_0}.$$

Therefore,

$$\mathbb{E}[X_n] \le \mathbb{E}[B_{\Delta,n}] + \left(1 + \frac{c_0}{\beta_0 - 1}\right) n^{-\alpha_0/\beta_0}. \tag{103}$$

Finally, Jensen's inequality gives

$$\begin{aligned}\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0, 0)\right| &\le \mathbb{E}\left[\left|\widehat{V}_n(s_0) - \widetilde{V}(s_0, 0)\right|\right] \\ &= \mathbb{E}[X_n].\end{aligned}$$

Combining this with (103) yields

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0, 0)\right| \le C_{\text{samp}} n^{-\alpha_0/\beta_0} + \mathbb{E}[B_{\Delta,n}],$$

for

$$C_{\text{samp}} := 1 + \frac{c_0}{\beta_0 - 1}.$$

Under the optimal tuning

$$\alpha_0/\beta_0 = 1/2,$$

we obtain

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0, 0)\right| \le C_{\text{samp}} n^{-1/2} + \mathbb{E}[B_{\Delta,n}]. \tag{104}$$

Since

$$B_{\Delta,n} = \frac{\delta_{Q,n}}{1-\gamma},$$

this proves the expected-error bound in terms of $\delta_{Q,n}$.

It remains to connect $\delta_{Q,n}$ to the stated cross-depth gap. Suppose

$$\delta_{Q,n} \le \gamma \Delta_{\text{CROSS}} \qquad \text{almost surely.}$$

Then

$$\mathbb{E}[B_{\Delta,n}] = \frac{1}{1-\gamma} \mathbb{E}[\delta_{Q,n}] \le \frac{\gamma}{1-\gamma} \Delta_{\text{CROSS}}.$$

Substituting into (104) gives

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0, 0)\right| \le C_{\text{samp}} n^{-1/2} + \frac{\gamma}{1-\gamma} \Delta_{\text{CROSS}}.$$

Thus

$$\left|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0, 0)\right| \le O(n^{-1/2}) + O(\Delta_{\text{CROSS}}),$$

as claimed. □

Remark 11 (Why Theorem 3 differs from Theorem 6). Theorem 6 cannot be copied verbatim to GS-Power-UCT-F. In the depth-augmented algorithm, each node is (s,h), every edge increases depth, and the graph is a DAG. Hence every node has a unique target $\widetilde{V}(s,h)$. In the full-merge algorithm, a node is keyed only by s, so the same estimator $\widehat{V}(s)$ may receive updates from several depths. Therefore the natural target is not a single depth-specific value, but the empirical full-merge fixed point $\bar{V}_n$ induced by the depth-mixture weights.

The bias term

$$B^V_{F,n}(s,h) = |\bar{V}_n(s) - \widetilde{V}(s,h)|$$

measures exactly the price of cross-depth merging. If there are no cross-depth transpositions, then every physical state is visited at only one depth, the weights $\omega_{n,h}(s,a)$ are point masses, and $\bar{V}_n(s) = \widetilde{V}(s,h)$. In that special case, $B^V_{F,n}(s,h) = 0$, and the full-merge theorem reduces to the same concentration statement as the depth-augmented theorem.

D.6 Proof of Theorem 4

Proof. Let

$$H_n \triangleq H(n).$$

By the triangle inequality,

$$\begin{aligned}\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)\right| \le \left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - \widetilde{V}_{H_n}(s_0,0)\right| \\ + \left|\widetilde{V}_{H_n}(s_0,0) - V^\star(s_0)\right|. \end{aligned} \tag{105}$$

The first term is exactly the fixed-horizon full-merge error controlled by Theorem 3, applied with horizon $H = H_n$. Therefore,

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - \widetilde{V}_{H_n}(s_0,0)\right| \le c(H_n)n^{-1/2} + C_\Delta\,\mathbb{E}\left[\Delta^{H_n,n}_{\text{CROSS}}\right]. \tag{106}$$

By Definition 3,

$$\Delta^n_{\text{CROSS}} = \Delta^{H(n),n}_{\text{CROSS}} = \Delta^{H_n,n}_{\text{CROSS}}.$$

Hence

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - \widetilde{V}_{H_n}(s_0,0)\right| \le c(H_n)n^{-1/2} + C_\Delta\,\mathbb{E}[\Delta^n_{\text{CROSS}}]. \tag{107}$$

For the second term, the finite-horizon truncation bound gives

$$\left|\widetilde{V}_{H_n}(s_0,0) - V^\star(s_0)\right| \le \gamma^{H_n}\|V^\star - V_0\|_\infty. \tag{108}$$

Combining the previous displays yields

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)\right| \le c(H_n)n^{-1/2} + C_\Delta\,\mathbb{E}[\Delta^n_{\text{CROSS}}] + \gamma^{H_n}\|V^\star - V_0\|_\infty. \tag{109}$$

Finally, by the adaptive-horizon choice

$$H_n = \left\lceil \frac{\log n}{2\log(1/\gamma)} \right\rceil,$$

we have

$$\gamma^{H_n} \le n^{-1/2}.$$

Therefore,

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)\right| \le c(H_n)n^{-1/2} + C_\Delta\,\mathbb{E}[\Delta^n_{\text{CROSS}}] + n^{-1/2}\|V^\star - V_0\|_\infty. \tag{110}$$

Thus, if

$$\mathbb{E}[\Delta^n_{\text{CROSS}}] \to 0 \qquad \text{and} \qquad c(H_n)n^{-1/2} \to 0,$$

then

$$\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] \to V^\star(s_0).$$

Moreover, if

$$\mathbb{E}[\Delta^n_{\text{CROSS}}] = O(n^{-1/2})$$

and $c(H_n)$ grows at most polylogarithmically in n, then

$$\left|\mathbb{E}[\widehat{V}_n^{H_n}(s_0)] - V^\star(s_0)\right| = O\left(\frac{c(H_n)}{\sqrt{n}}\right).$$

□

E Experimental Details

E.1 Evaluation Protocol

To evaluate the performance and scalability of each algorithm, we define a set of simulation budgets (number of iterations) tailored to the complexity of each environment. For every environment, each algorithm is evaluated across these simulation checkpoints to observe its performance profile.

Our evaluation follows a replanning paradigm. Specifically, at each time step of an episode, the agent executes a complete search process using its allotted simulation budget to determine the optimal action. This process is repeated for every step until a terminal state is reached. To ensure statistical significance and account for the stochastic nature of Monte Carlo-based methods, we conduct 1,000 independent runs for each combination of algorithm, environment, and simulation budget, except for GBOP, which uses 500 runs. The final performance is reported as the mean cumulative reward (or other environment-specific metric) over the corresponding evaluation runs.

E.2 Hyperparameter Selection

To ensure a fair comparison, we perform an extensive hyperparameter sweep for both the proposed method and the baselines using a grid search strategy. The search space for each algorithm is defined based on literature-standard ranges and preliminary runs.

The tuning process is conducted as follows:

- For each environment and algorithm, we evaluate candidate hyperparameter sets using 300 independent runs.
- These runs are performed across all predefined simulation budget checkpoints to ensure the robustness of the selected parameters.
- **Selection Criterion:** The optimal hyperparameter configuration is chosen based on the highest aggregate mean reward, calculated by averaging the performance across all simulation budgets within a specific environment.

By selecting parameters that perform well across different simulation budgets, we aim to identify configurations that are not only high-performing but also stable under varying computational constraints.

E.3 Benchmark Environments

E.3.1 FrozenLake

Environment Description: The Frozen Grid environment consists of an 8×8 grid where the agent starts at coordinate $(0, 0)$ and aims to reach the goal state at $(7, 7)$. The state space is defined by the agent's (x, y) coordinates. The grid cells are categorized into Frozen (F), Holes (H), Start (S), and Goal (G). The action space consists of four discrete movements: *left*, *down*, *right*, and *up*. To introduce stochasticity (slippery mode), transitions are non-deterministic: the executed action results in the intended direction with a probability of $1/3$, or deviates to a perpendicular direction (left turn or right turn) with a probability of $1/3$ each. Boundary collisions result in the agent remaining in its current cell. The episode terminates if the agent reaches the goal, falls into a hole, or exhausts the maximum horizon limit of $H = 200$. The reward function is strictly sparse, granting a reward of 1 only upon reaching the goal, and 0 for all other transitions.

Selected Hyperparameters: UCT($\epsilon = 1.5$), MENTS(temperature=0.25, $\epsilon = 0.5$), Power-UCT($C = 1.5$, $p = 2.0$), GS-Power-UCT($C = 0.5$, $p = \infty$), GS-Power-UCT-F$^+$($C = 1.25$, $p = \infty$).

E.3.2 Passenger Grid

Environment Description: The Passenger Grid is formulated on a 7×6 layout. The state is represented as a tuple (x, y, M), where M is a bitmask tracking the pickup status of passengers. The agent starts at $(0, 0)$ and must deliver passengers to the drop-off goal at $(6, 0)$. Three passengers are initially located at $(1, 2)$, $(0, 5)$, and $(6, 4)$. The i-th bit of M flips to 1 when the agent occupies passenger i's initial cell. Actions include *left*, *down*, *right*, and *up*. The environment operates in a slippery mode where the effective action is sampled from $\{\text{turn-left}, \text{intended}, \text{turn-right}\}$ with probabilities $\{0.25, 0.5, 0.25\}$, respectively. Boundary collisions keep the agent stationary. Each transition increments the time step, and episodes terminate either upon reaching the drop-off cell or at horizon $H = 70$. The reward is delayed until the goal is reached and depends on the total number of collected passengers: 0, 1, 3, or 7 for collecting 0, 1, 2, or 3 passengers, respectively. All other transitions yield a reward of 0.

Selected Hyperparameters: UCT($\epsilon = 1.5$), MENTS(temperature=0.01, $\epsilon = 0.5$), Power-UCT($C = 1.5$, $p = 2.5$), GBOP($\epsilon = 0.001$, $\gamma = 0.99$), GS-Power-UCT($C = 0.5$, $p = \infty$), GS-Power-UCT-F$^+$($C = 0.5$, $p = 4.0$).

E.3.3 Factored River Swim

Environment Description: The Factored River Swim environment comprises 3 independent river-swim chains. The joint state captures the position of an agent on each chain, $P \in \{0, \ldots, 4\}^3$. All rivers initialize at position 0, with 3 being the target goal. A joint action is represented as a 3-bit vector (8 possible actions), where the i-th bit indicates the behavior for river i: 0 for resting/swimming downstream (left) and 1 for swimming upstream (right). Action 0 deterministically moves the agent one step left (bounded at 0). Action 1 introduces stochastic transitions: at position 0, the agent stays with probability 0.4 and moves right with 0.6; at the goal (4), it falls back to 3 with probability 0.4 and stays with 0.6; at any intermediate state, it moves left (0.05), stays (0.60), or moves right (0.35). Episodes terminate at $H = 35$. Rewards are evaluated prior to position updates. For each river, selecting action 0 at position 0 yields 0.1, while selecting action 1 at position 7 yields 1.0. A synergistic bonus of 4.0 is granted if all rivers are simultaneously at position 7 and all action bits are set to 1. The total aggregated reward is normalized by a factor of 6.

Selected Hyperparameters: UCT($\epsilon = 0.5$), MENTS(temperature=0.99, $\epsilon = 0.5$), Power-UCT($C = 1.5$, $p = 1.0$), GBOP($\epsilon = 0.001$, $\gamma = 0.99$), GS-Power-UCT($C = 0.5$, $p = \infty$), GS-Power-UCT-F$^+$($C = 1.5$, $p = \infty$).

E.3.4 SysAdmin Ring

Environment Description: The SysAdmin Ring models a network of 20 interconnected computers forming a ring topology. The state is represented by an *alive_mask*, where each bit denotes the running status of a specific machine. Initially, all computers are down (*alive_mask* $= 0$). The agent can choose from 21 actions: either reboot a specific computer $i \in \{0, \ldots, 19\}$ or remain idle. Rebooting a computer guarantees it will be running in the subsequent state. For all other machines, the status updates stochastically based on their current state and that of their predecessor in the ring. The running probabilities are determined as follows: 0.0238 if both are down; 0.0475 if the predecessor is running but the current is down; 0.525 if the predecessor is down but the current is running; and 0.95 if both are running. Episodes terminate deterministically at $t = 50$. The environment provides a dense step-wise reward corresponding to the fraction of running machines.

Selected Hyperparameters: UCT($\epsilon = 1.5$), MENTS(temperature=0.25, $\epsilon = 0.5$), Power-UCT($C = 0.5$, $p = 3.0$), GS-Power-UCT($C = 0.5$, $p = 2.5$), GBOP($\epsilon = 0.001$, $\gamma = 0.99$), GS-Power-UCT-F$^+$($C = 0.75$, $p = 2.5$).

E.3.5 FourRooms

Environment Description: FourRooms is an 11×11 grid partitioned by a vertical wall at $x = 5$ and a horizontal wall at $y = 5$. Communication between rooms is facilitated by four door cells, one randomly placed on each of the four wall segments during reset. The agent's start and goal positions are uniformly sampled from non-wall and non-door cells. The agent executes four directional actions, subject to slippery dynamics identical to the Passenger Grid: effective actions are sampled from $\{\text{turn-left}, \text{intended}, \text{turn-right}\}$ with probabilities $\{0.25, 0.5, 0.25\}$. Collisions with walls or boundaries leave the agent in place. Episodes terminate upon reaching the target goal or hitting the time limit $H = 50$. The reward is 0 at all steps except when the goal is reached, where a time-discounted reward is issued: $R = 1 - 0.9 \times (t/50)$, incentivizing faster navigation.

Selected Hyperparameters: UCT($\epsilon = 1.0$), MENTS(temperature=0.01, $\epsilon = 0.5$), Power-UCT($C = 1.0$, $p = 2.0$), GS-Power-UCT($C = 0.5$, $p = \infty$), GBOP($\epsilon = 0.001$, $\gamma = 0.99$), GS-Power-UCT-F$^+$($C = 0.5$, $p = 4.0$).

E.4 Experimental Results

The following tables report reward as the mean $\pm$ twice the standard deviation over 1,000 evaluation runs for each algorithm and simulation budget (except GBOP: 500 runs). Each table is split into two budget panels for readability; the column headings give the number of simulations per decision. P-UCT denotes Stochastic-Power-UCT.

E.5 Runtime and memory analysis

For a fixed horizon H, each search depth requires an expected-$O(1)$ hash lookup and $O(K)$ action-selection and backup work, giving $O(HK)$ time per simulation, the same asymptotic order as tree-based Stochastic-Power-UCT. For depth-augmented GS-Power-UCT, Theorem 2 implies that the node and edge counts under the same realised trajectories do not exceed those of the unrolled tree. This representation statement does not apply to independently run planners or to full merging. The graph additionally stores a hash table and adjacency lists, requiring $O(|\mathcal{N}| + |\mathcal{E}|)$

Table 5: FrozenLake

Algorithm	16	32	64	128
UCT	0.002±0.089	0.004±0.126	0.006±0.155	0.011±0.209
P-UCT	0.003±0.109	0.006±0.155	0.004±0.126	0.012±0.218
MENTS	0.002±0.089	0.011±0.209	0.016±0.251	0.067±0.500
GBOP	0.000±0.000	0.000±0.000	0.060±0.011	0.198±0.018
GS-Power-UCT	0.051±0.440	0.140±0.694	0.227±0.838	0.314±0.929
GS-Power-UCT-F$^+$	0.105±0.613	0.139±0.692	0.183±0.774	0.208±0.812
Algorithm	256	512	1024	2048
UCT	0.016±0.251	0.021±0.287	0.047±0.423	0.064±0.490
P-UCT	0.020±0.280	0.014±0.235	0.047±0.423	0.061±0.479
MENTS	0.127±0.666	0.156±0.726	0.209±0.814	0.257±0.874
GBOP	0.236±0.019	0.262±0.020	0.246±0.019	0.294±0.020
GS-Power-UCT	0.395±0.978	0.480±1.000	0.527±0.999	0.560±0.993
GS-Power-UCT-F$^+$	0.276±0.894	0.353±0.956	0.413±0.985	0.478±1.000

Table 6: Passenger Grid

Algorithm	16	32	64	128
UCT	0.685±2.170	0.828±2.430	1.053±2.853	1.413±3.114
P-UCT	0.661±2.216	0.861±2.598	1.220±3.143	1.580±3.249
MENTS	0.587±2.211	0.739±2.571	0.911±2.843	1.298±3.461
GBOP	0.000±0.000	0.000±0.000	0.000±0.000	0.046±0.013
GS-Power-UCT	1.877±4.662	3.062±5.607	3.919±5.883	4.544±5.888
GS-Power-UCT-F$^+$	1.253±2.552	1.324±2.516	1.413±2.810	1.452±2.831
Algorithm	256	512	1024	2048
UCT	1.668±3.117	2.093±3.206	2.432±3.348	2.586±3.167
P-UCT	1.966±3.587	2.524±3.815	2.955±3.976	3.366±4.138
MENTS	1.679±3.710	2.091±4.035	2.736±4.448	3.543±4.761
GBOP	1.570±0.073	3.756±0.106	3.452±0.096	3.546±0.092
GS-Power-UCT	5.178±5.507	5.363±5.441	5.801±4.870	6.134±4.294
GS-Power-UCT-F$^+$	1.574±2.951	1.684±2.957	1.847±3.170	1.993±3.364

Table 7: Factored River Swim

Algorithm	16	32	64	128
UCT	1.319±0.960	1.344±0.906	1.597±1.304	1.809±1.542
P-UCT	1.654±1.497	1.999±1.778	2.348±1.946	2.666±2.147
MENTS	1.121±0.270	1.352±0.189	1.535±0.145	1.645±0.107
GBOP	1.750±0.000	1.749±0.000	1.745±0.006	2.165±0.032
GS-Power-UCT	2.552±1.434	3.259±1.784	4.353±2.166	5.865±3.522
GS-Power-UCT-F$^+$	5.784±3.378	6.649±3.727	6.913±3.718	6.990±3.868
Algorithm	256	512	1024	2048
UCT	1.995±1.577	2.286±1.722	2.440±1.836	2.437±1.721
P-UCT	3.010±2.433	3.186±2.667	2.968±2.643	2.541±2.144
MENTS	1.720±0.058	1.749±0.011	1.750±0.000	1.750±0.000
GBOP	2.583±0.043	3.232±0.059	4.261±0.076	5.290±0.073
GS-Power-UCT	6.754±3.664	6.977±3.586	7.048±3.690	7.044±3.687
GS-Power-UCT-F$^+$	6.847±3.953	6.496±4.194	5.808±4.227	5.144±4.102

Table 8: SysAdmin Ring

Algorithm	16	32	64	128
UCT	8.574±2.692	8.687±2.835	8.883±3.039	8.980±3.105
P-UCT	8.575±2.690	8.731±2.885	9.078±3.030	9.479±3.419
MENTS	8.542±2.734	8.630±2.894	8.652±2.856	8.799±2.966
GBOP	9.037±0.067	8.452±0.061	8.108±0.056	8.090±0.054
GS-Power-UCT	8.662±2.677	9.018±2.947	9.440±3.233	10.159±3.906
GS-Power-UCT-F$^+$	8.867±2.763	9.334±3.049	9.757±3.210	10.656±3.749
Algorithm	256	512	1024	2048
UCT	9.180±3.372	9.828±3.986	10.722±4.249	12.139±5.583
P-UCT	10.197±4.009	11.168±4.741	12.872±5.954	14.810±6.413
MENTS	8.821±2.985	8.888±3.116	9.140±3.346	9.484±3.514
GBOP	8.029±0.052	8.159±0.055	8.414±0.053	8.580±0.055
GS-Power-UCT	11.307±4.602	13.264±5.627	15.601±6.075	17.205±6.126
GS-Power-UCT-F$^+$	12.218±4.518	14.107±5.398	15.892±5.979	16.794±6.192

Table 9: FourRooms

Algorithm	16	32	64	128
UCT	0.176±0.618	0.200±0.628	0.218±0.660	0.253±0.672
P-UCT	0.165±0.602	0.208±0.656	0.245±0.686	0.281±0.703
MENTS	0.155±0.591	0.185±0.627	0.218±0.655	0.241±0.668
GBOP	0.163±0.014	0.209±0.015	0.227±0.015	0.282±0.015
GS-Power-UCT	0.338±0.654	0.420±0.626	0.478±0.596	0.521±0.551
GS-Power-UCT-F$^+$	0.396±0.676	0.443±0.662	0.469±0.644	0.484±0.622
Algorithm	256	512	1024	2048
UCT	0.290±0.689	0.347±0.706	0.399±0.694	0.439±0.675
P-UCT	0.328±0.717	0.381±0.707	0.407±0.698	0.457±0.672
MENTS	0.302±0.685	0.352±0.694	0.390±0.688	0.434±0.671
GBOP	0.383±0.014	0.409±0.014	0.440±0.013	0.327±0.015
GS-Power-UCT	0.545±0.530	0.554±0.525	0.566±0.512	0.572±0.502
GS-Power-UCT-F$^+$	0.494±0.618	0.501±0.604	0.509±0.594	0.513±0.586

auxiliary storage. A finite discrete MDP stores at most $|\mathcal{S}|(H+1)$ depth-augmented nodes or $|\mathcal{S}|$ full-merge nodes; before saturation, both quantities are budget-dependent.

To illustrate implementation overhead, we compare the proposed algorithms with Stochastic-Power-UCT. GS-Power-UCT adds state merging. For GS-Power-UCT-F$^+$, each reported budget n uses the fixed horizon $H(n)$ for all n simulations in that search. We report three representative budgets for brevity. Each budget is reported in two rows: average wall-clock time per decision (episode time divided by steps, averaged over 1,000 episodes), and average expanded-node count over all steps.

Runtime and memory interpretation: The largest observed positive runtime difference is 15.5% (Table 10); some measurements are lower than the tree baseline. Expanded-node counts indicate retained graph size and are not direct measurements of RAM use. The proposed methods add hash-table and graph-bookkeeping storage, while node-count reduction can be substantial when transpositions are common. Without transpositions, node counts approach those of the tree representation, with additional bookkeeping constants.

Table 10: Wall-clock time per decision and average expanded-node count across environments and simulation budgets. Node counts are a retained-graph-size proxy, not memory measurements.

Environment	Sims	Metric	Stochastic-Power-UCT	GS-Power-UCT	GS-Power-UCT-F$^+$
FourRooms	16	Time	0.0146 ms	0.0155 ms (+6.2%)	0.0146 ms (+0.0%)
		Nodes	17.0	8.9 (-47.6%)	8.3 (-51.2%)
FourRooms	256	Time	0.2513 ms	0.2771 ms (+10.3%)	0.2875 ms (+14.4%)
		Nodes	257.0	76.9 (-70.1%)	51.9 (-79.8%)
FourRooms	2048	Time	4.6116 ms	5.3265 ms (+15.5%)	5.2128 ms (+13.0%)
		Nodes	1971.7	80.8 (-95.9%)	54.1 (-97.3%)
Passenger Grid	16	Time	0.0199 ms	0.0199 ms (+0.0%)	0.0201 ms (+1.0%)
		Nodes	17.0	9.1 (-46.5%)	8.5 (-50.0%)
Passenger Grid	256	Time	0.3155 ms	0.3161 ms (+0.2%)	0.3273 ms (+3.7%)
		Nodes	257.0	75.1 (-70.8%)	51.9 (-79.8%)
Passenger Grid	2048	Time	3.9591 ms	4.1618 ms (+5.1%)	4.2138 ms (+6.4%)
		Nodes	1732.4	81.5 (-95.3%)	56.8 (-96.7%)
Factored River Swim	16	Time	0.0182 ms	0.0185 ms (+1.6%)	0.0185 ms (+1.6%)
		Nodes	17.0	17.0 (-0.0%)	17.0 (-0.0%)
Factored River Swim	256	Time	0.3205 ms	0.3311 ms (+3.3%)	0.3336 ms (+4.1%)
		Nodes	257.0	174.5 (-32.1%)	161.2 (-37.3%)
Factored River Swim	2048	Time	5.5256 ms	5.6520 ms (+2.3%)	5.6782 ms (+2.8%)
		Nodes	2049.0	702.2 (-65.7%)	463.7 (-77.4%)
SysAdmin Ring	16	Time	0.0215 ms	0.0213 ms (-0.9%)	0.0213 ms (-0.9%)
		Nodes	17.0	17.0 (-0.0%)	17.0 (-0.0%)
SysAdmin Ring	256	Time	0.3730 ms	0.3748 ms (+0.5%)	0.3758 ms (+0.8%)
		Nodes	257.0	241.7 (-6.0%)	234.3 (-8.8%)
SysAdmin Ring	2048	Time	6.0448 ms	6.1197 ms (+1.2%)	6.1136 ms (+1.1%)
		Nodes	2049.0	1795.9 (-12.4%)	1663.7 (-18.8%)
FrozenLake	16	Time	0.1690 ms	0.1669 ms (-1.2%)	0.1665 ms (-1.5%)
		Nodes	17.0	8.7 (-48.8%)	8.1 (-52.4%)
FrozenLake	256	Time	2.9186 ms	2.9782 ms (+2.0%)	2.5900 ms (-11.3%)
		Nodes	257.0	74.9 (-70.9%)	49.3 (-80.8%)
FrozenLake	2048	Time	30.7731 ms	31.5220 ms (+2.4%)	30.0416 ms (-2.4%)
		Nodes	2008.7	78.0 (-96.1%)	51.5 (-97.4%)